

上海科技大学
ShanghaiTech University

Color

CS172 Computer Vision I

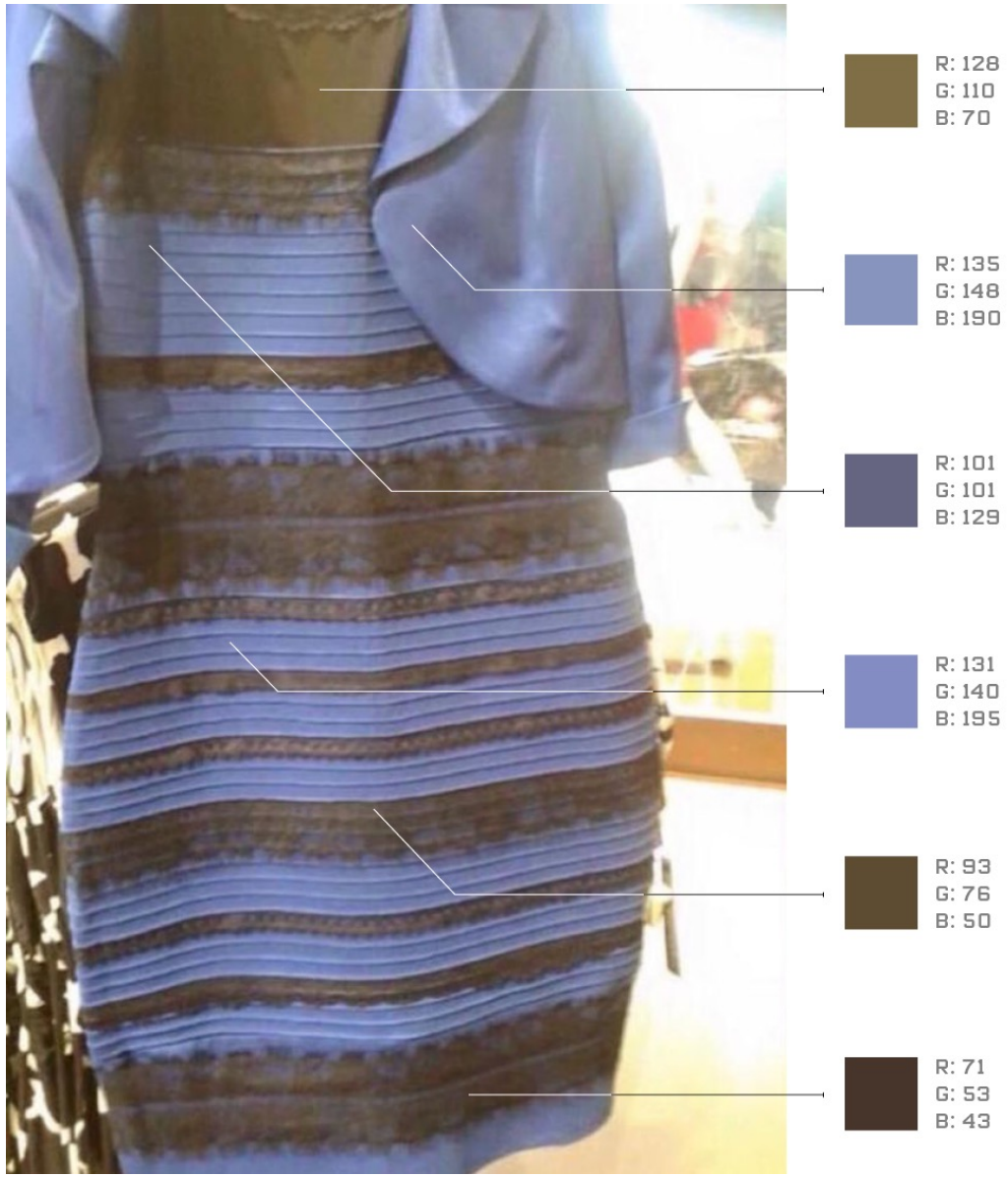
Instructor: Jiayuan Gu

What’s the color of dress?



http://www.wired.com/2015/02/science-one-agrees-color-dress/

How to represent color in computer?



Color and Light

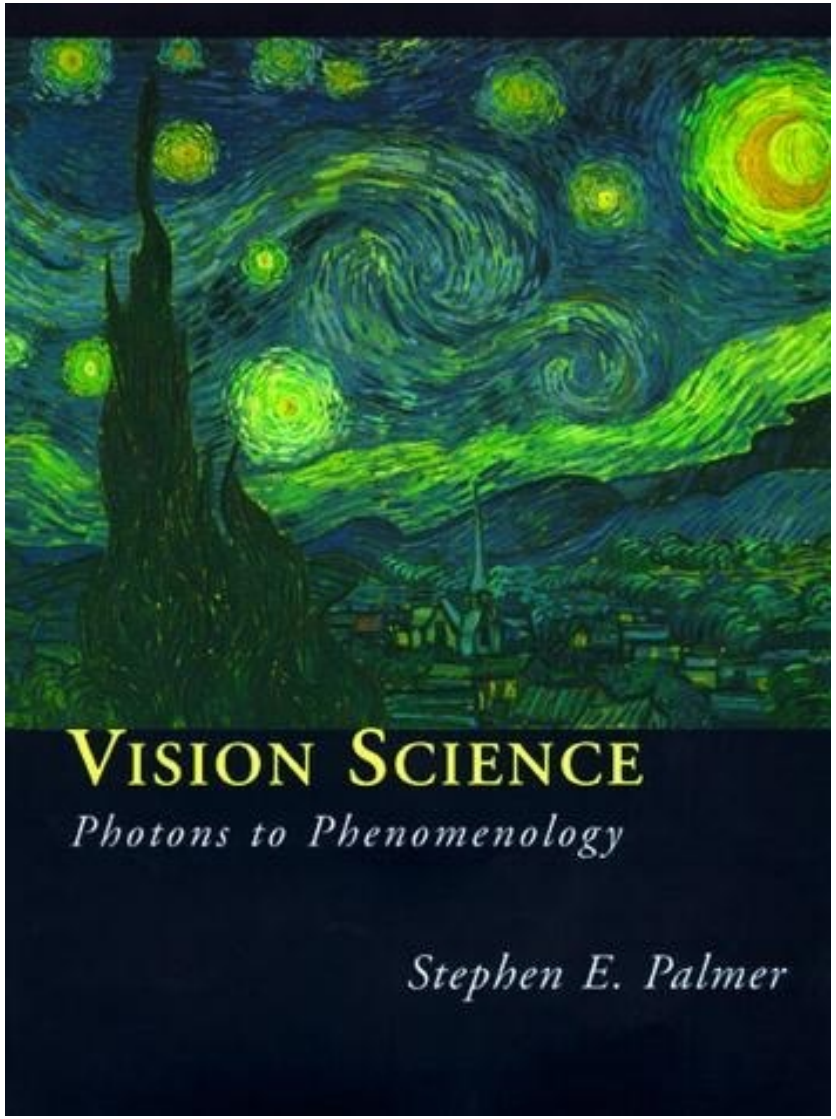
- **Color of light** arriving at camera depends on
  - Spectral reflectance of the surface light is leaving
  - Spectral radiance of light falling on that patch

- **Color perceived** depends on
  - Physics of light
  - Visual system receptors
  - Brain processing

What is color?

- Color is the result of interaction between physical light in the environment and our visual system

- Color is a psychological property of our visual experiences when we look at objects and lights, *not* a physical property of those objects or lights
  -- S. Palmer, *Vision Science: Photons to Phenomenology*



Agenda

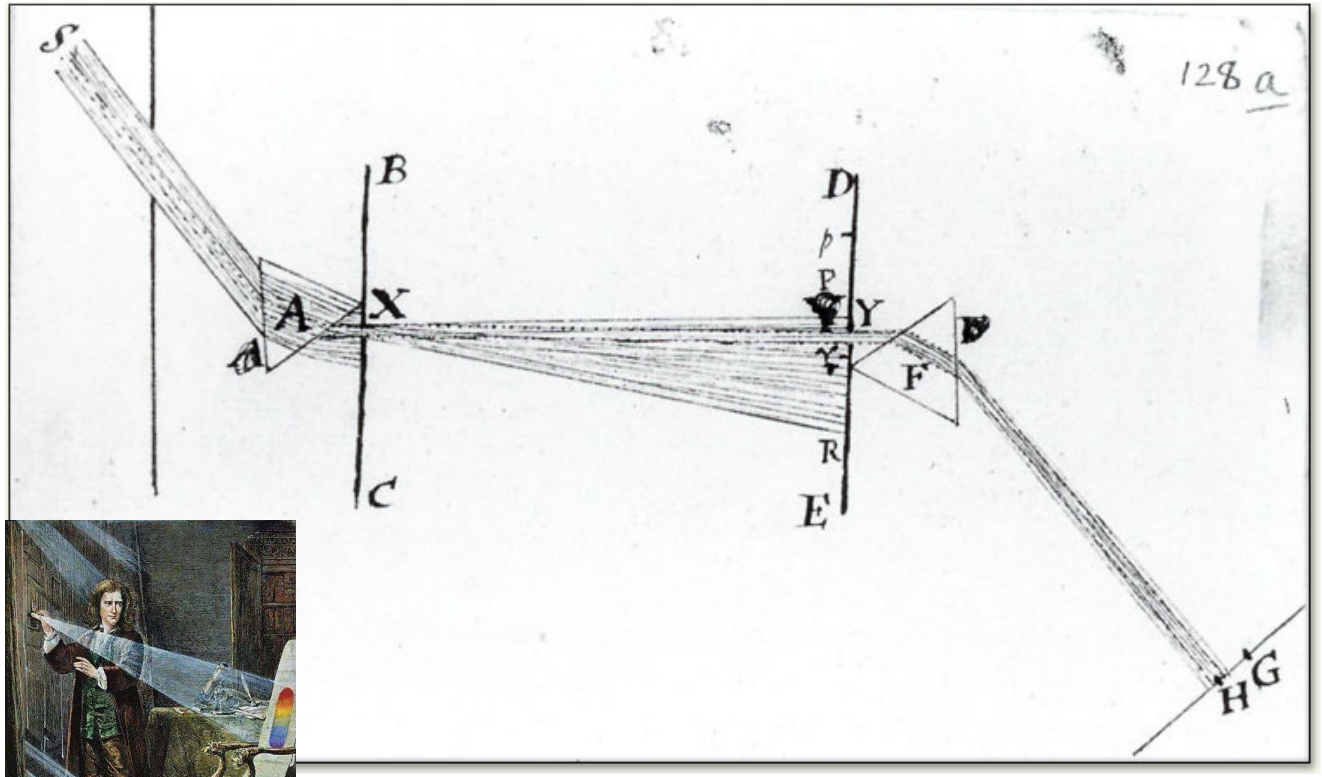
- Roadmap
  - Spectrum
  - Human color perception
  - Color space
  - Applications

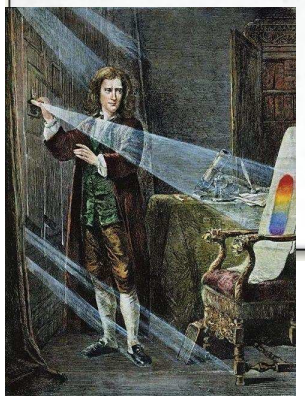
- Objectives
  - explain spectral image formation
  - explain why three measurements can represent color
  - distinguish RGB/XYZ/HSV

Spectrum

Physics of light

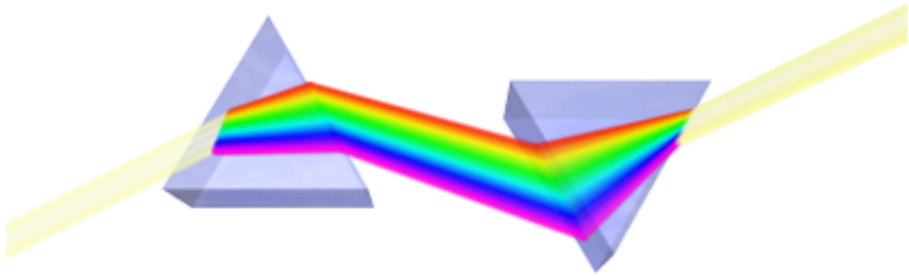
Electromagnetic Spectrum





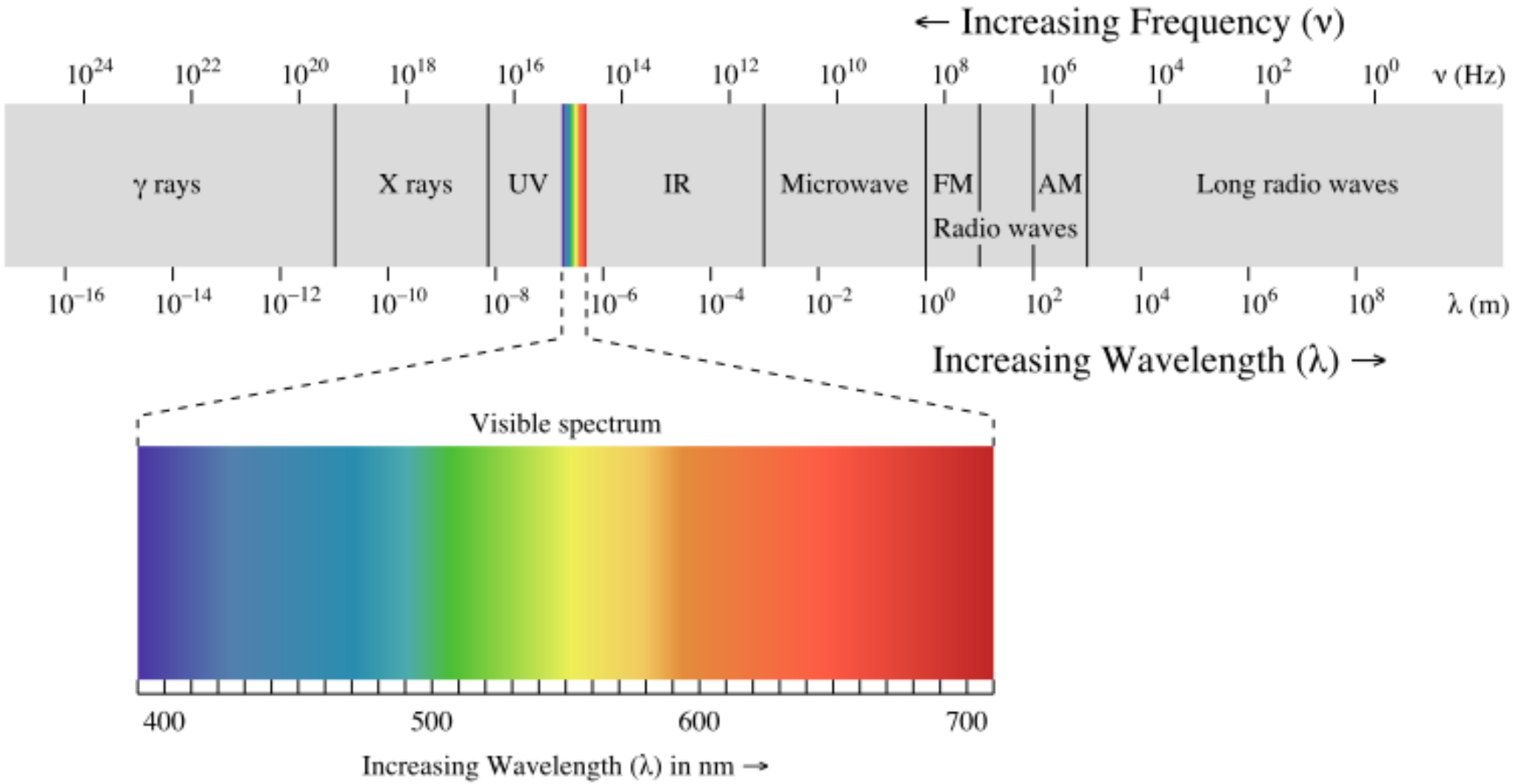
Newton, 1665

Image from http://micro.magnet.fsu.edu/

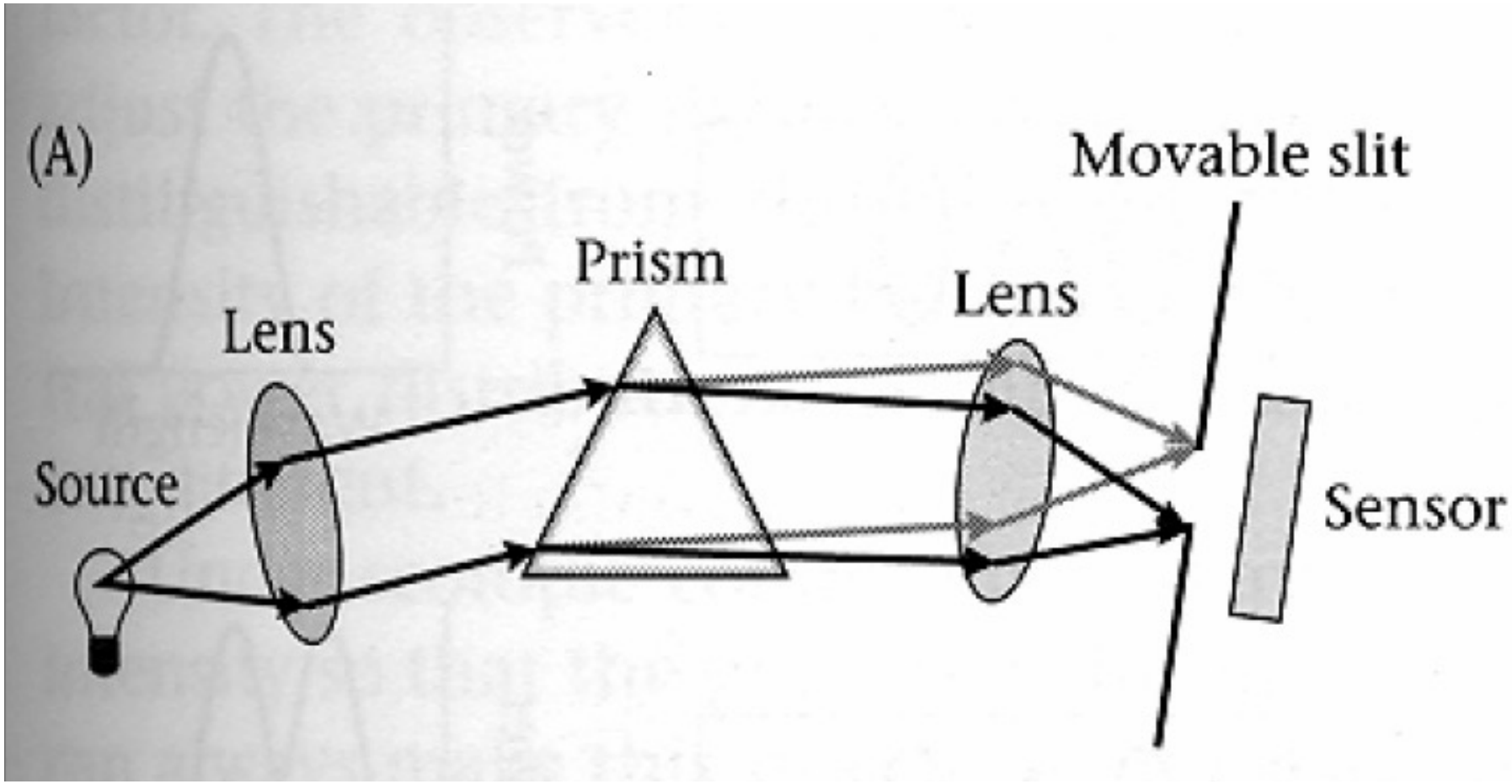


White light: composed of about equal energy in all wavelengths of the visible spectrum

Visible Spectrum



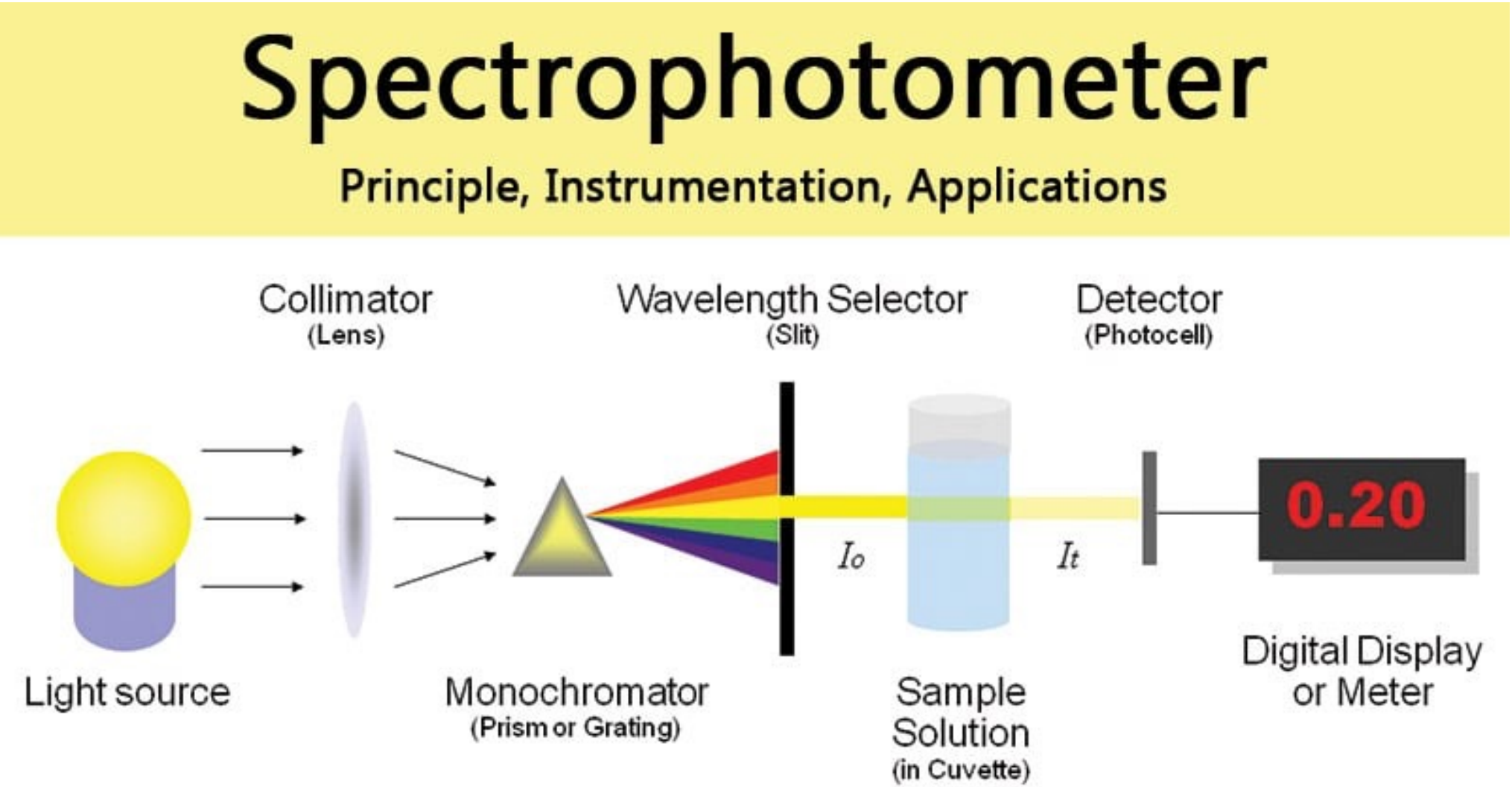
Measuring Spectra



Spectroradiometer: separate input light into its different wavelengths, and measure the energy at each.

Foundations of Vision, B. Wandell

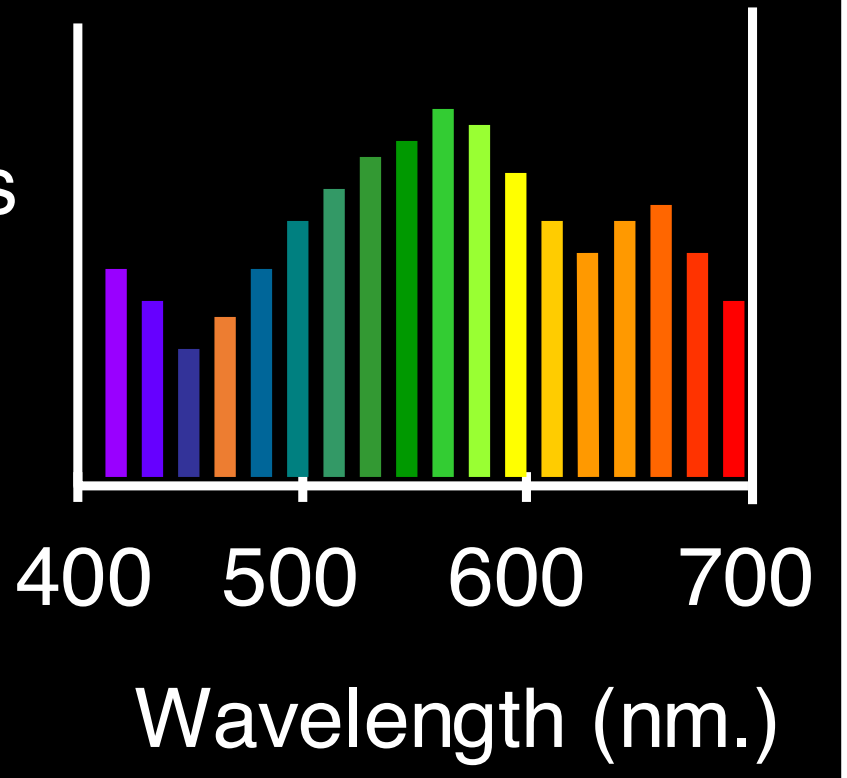
Measuring Spectra



https://microbenotes.com/spectrophotometer-principle-instrumentation-applications/

Spectrum of light

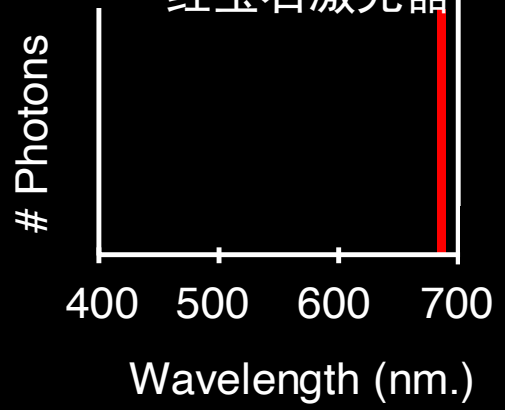
Any source of light can be completely described physically by its spectrum: the amount of energy emitted (per time unit) at each wavelength 400 - 700 nm.



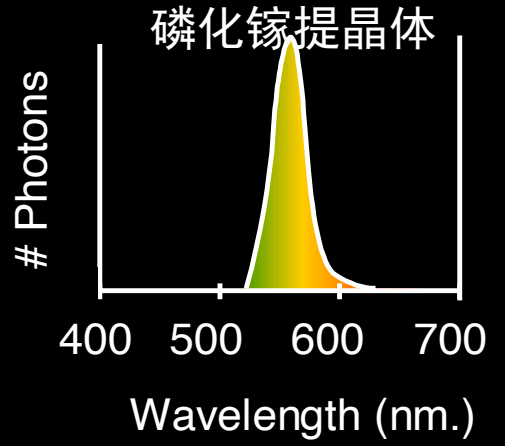
© Stephen E. Palmer, 2002

Spectra of Light Sources

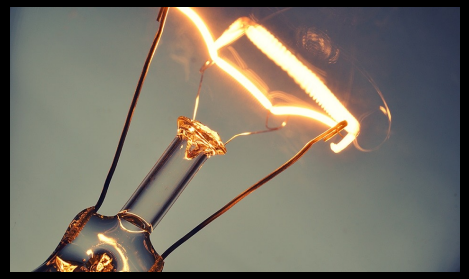
A. Ruby Laser
红宝石激光器



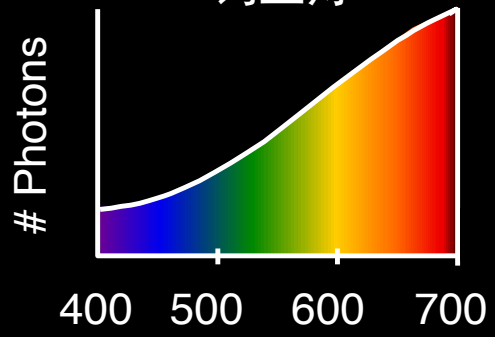
B. Gallium Phosphide Crystal
磷化镓提晶体



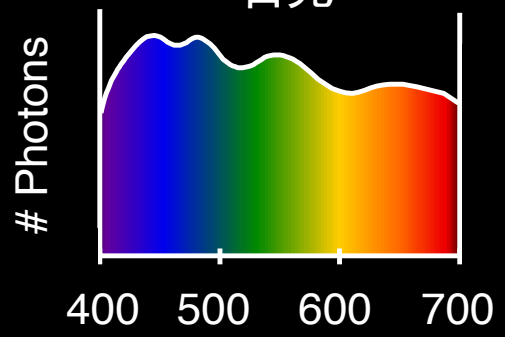




C. Tungsten Lightbulb
钨丝灯



D. Normal Daylight
日光



© Stephen E. Palmer, 2002

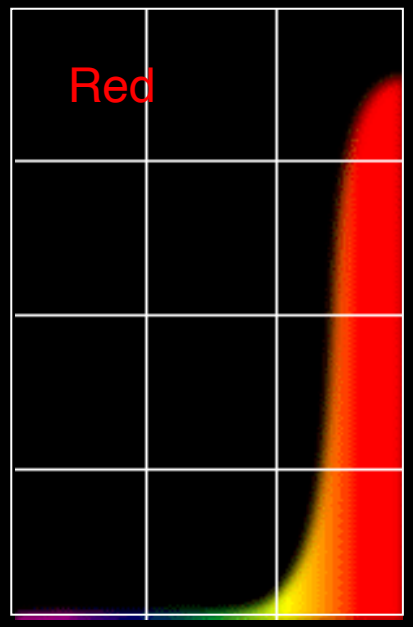
Reflectance Spectra

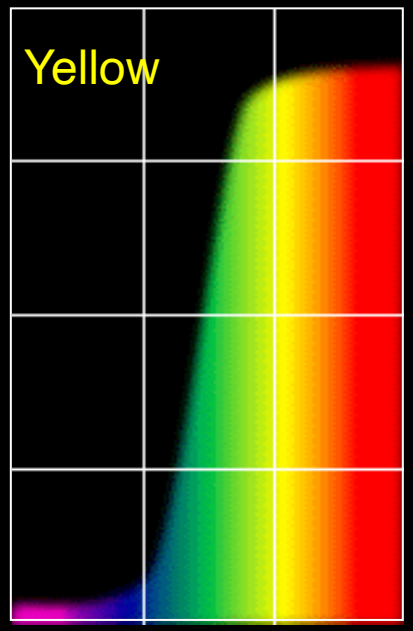


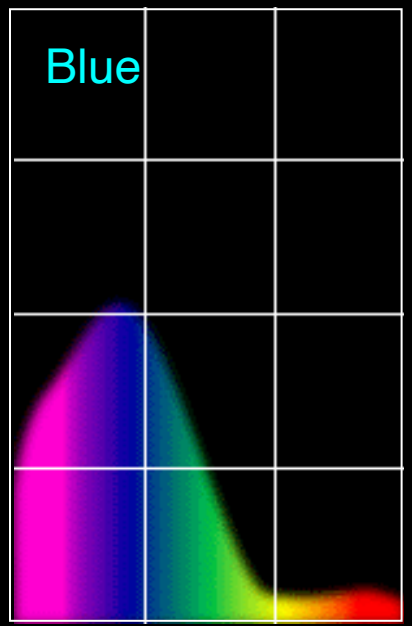


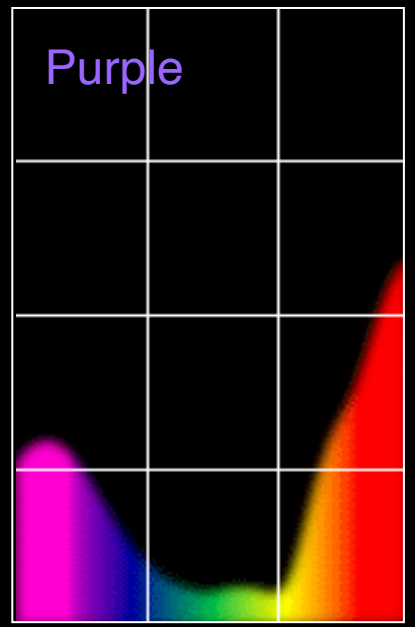










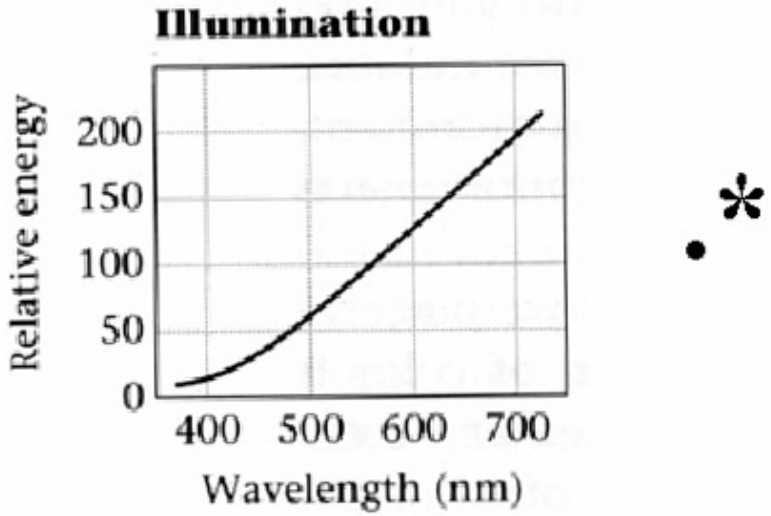


© Stephen E. Palmer, 2002

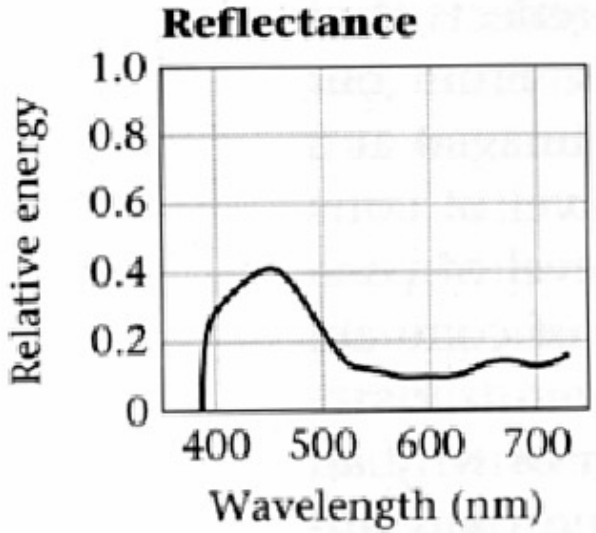
From Light and Surfaces to Spectra



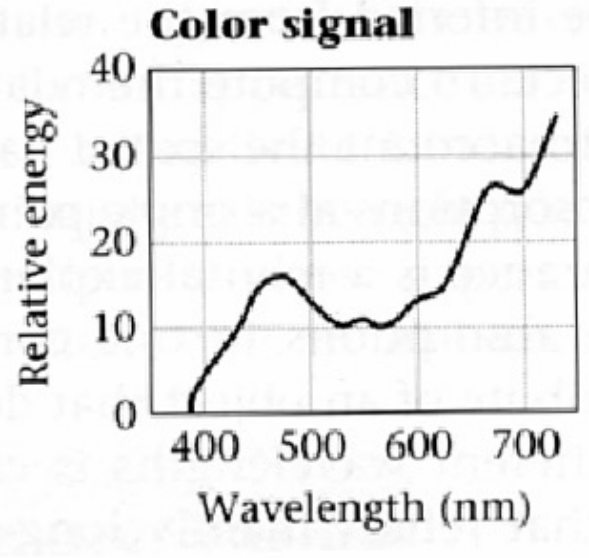
Reflected color is the result of interaction of light source spectrum with surface reflectance



•*



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Human Color Perception

From a Spectrum to Three Measurements

Tri-Chromatic Theory



additive primary colors
red, green, blue

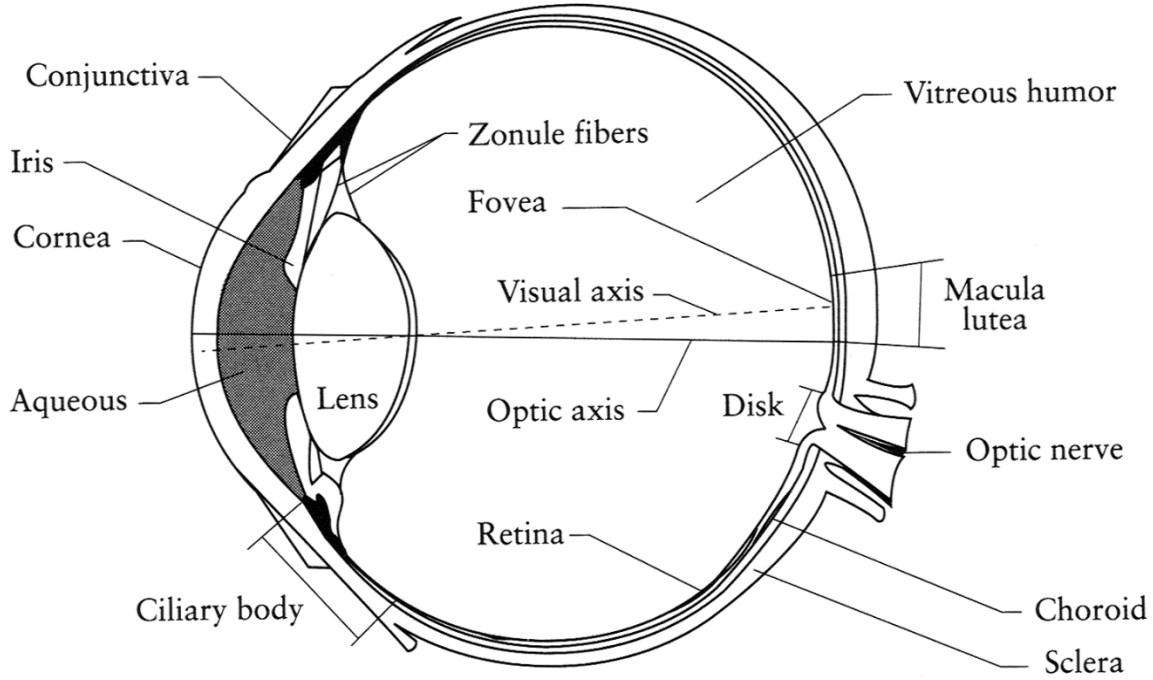


subtractive primary colors
cyan, magenta, yellow

If light is a spectrum, why can mixing two primary colors lead to a third color?

Are the wavelengths somehow mixed up to produce a new wavelength?

The Eye



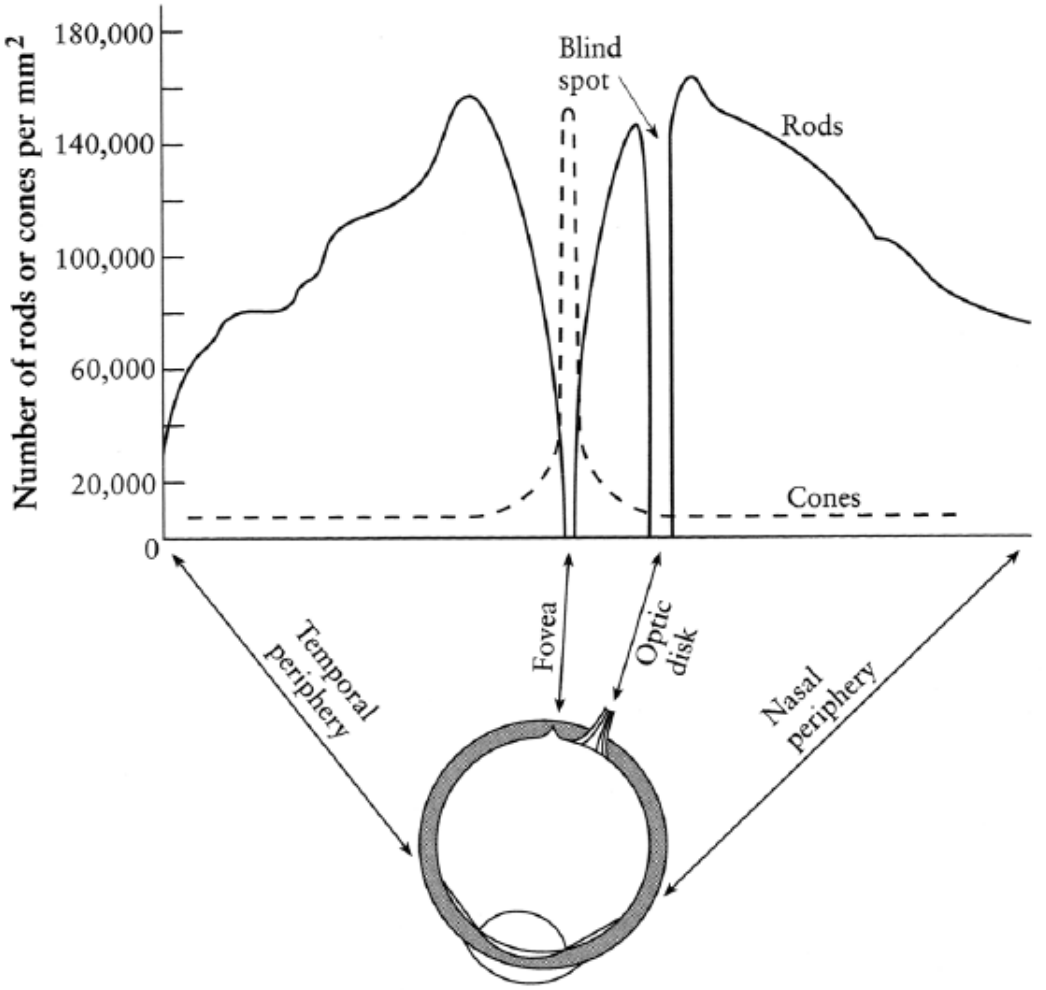
The human eye is a camera!

- **Iris** 虹膜 - colored annulus with radial muscles
- **Pupil** 瞳孔 - the hole (aperture) whose size is controlled by the iris
- **Lens** 晶状体 - changes shape by using ciliary muscles (to focus on objects at different distances)
- **Retina** 视网膜 - photoreceptor cells 感光器

Credit to Steve Seitz

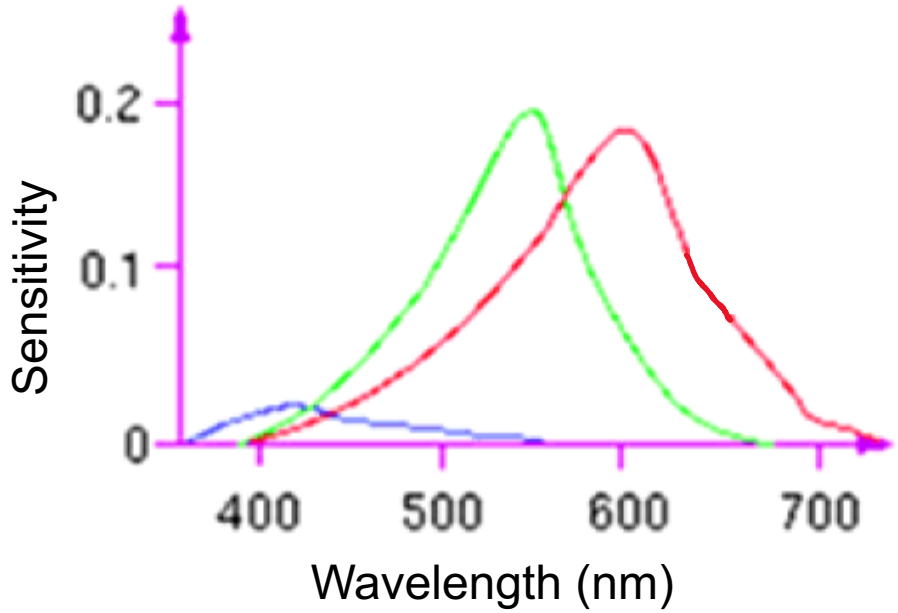
Two Types of Light-Sensitive Receptors

- Cones
  - Cone-shaped
  - Operated in strong light
  - Color vision
- Rods
  - Rod-shaped
  - Operated at night
  - Gray-scale vision



Cones

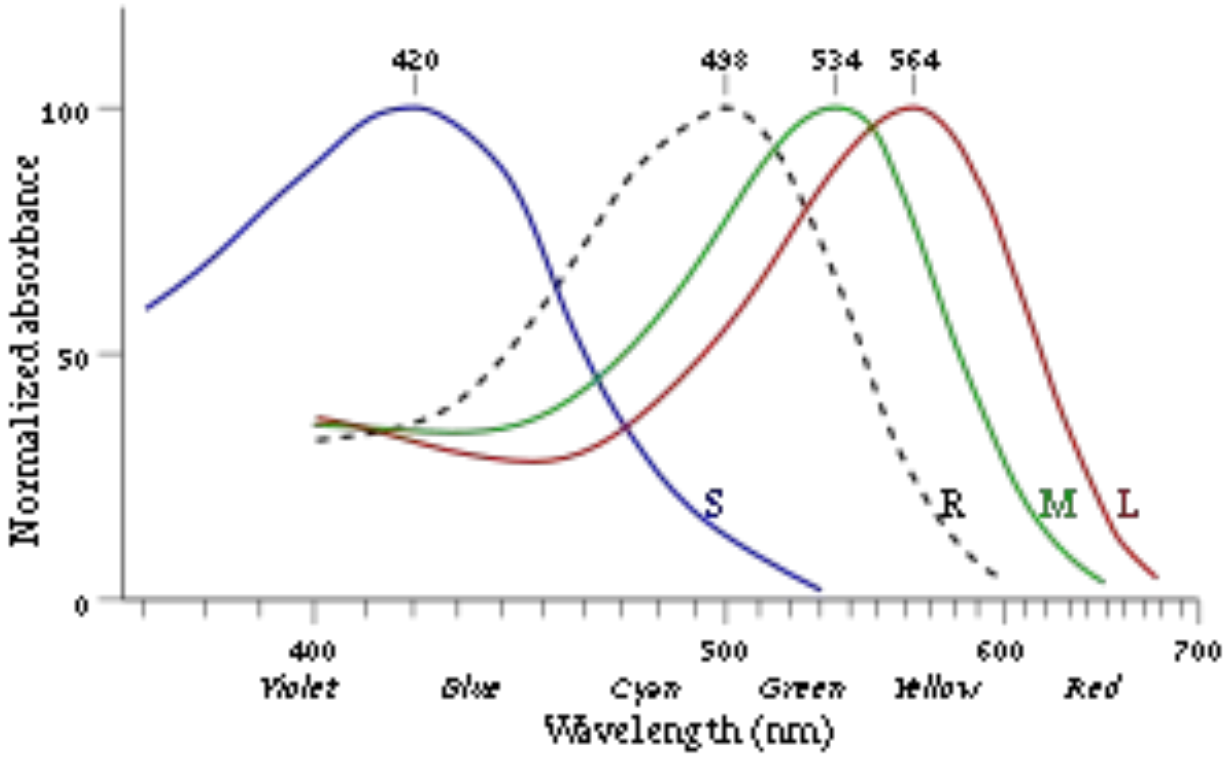
- React only to some wavelengths, with different sensitivity (light fraction absorbed)
- Brain fuses responses from local neighborhood of several cones for perceived color
- Sensitivities vary per person, and with age
- Color blindness: deficiency in at least one type of cone



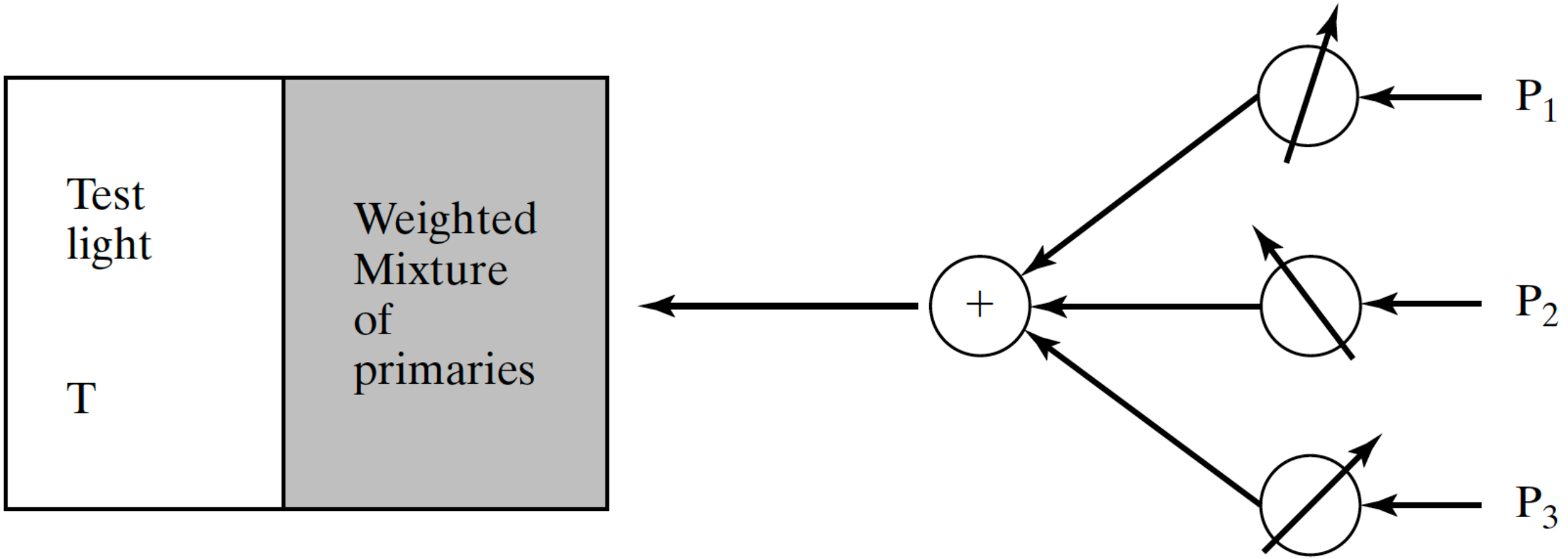
Human Color Receptors

- Long (red), Medium (green), and Short (blue) cones, plus intensity rods

- Fun Facts
  - Gene differences for color-blindness: “M” and “L” on the X-chromosome
  - Animals can have 1 (night animals), 2 (e.g., dogs), 4 (fish, birds), 5 (pigeons, some reptiles / amphibians), or even 12 (mantis shrimp) types of cones



Color Matching

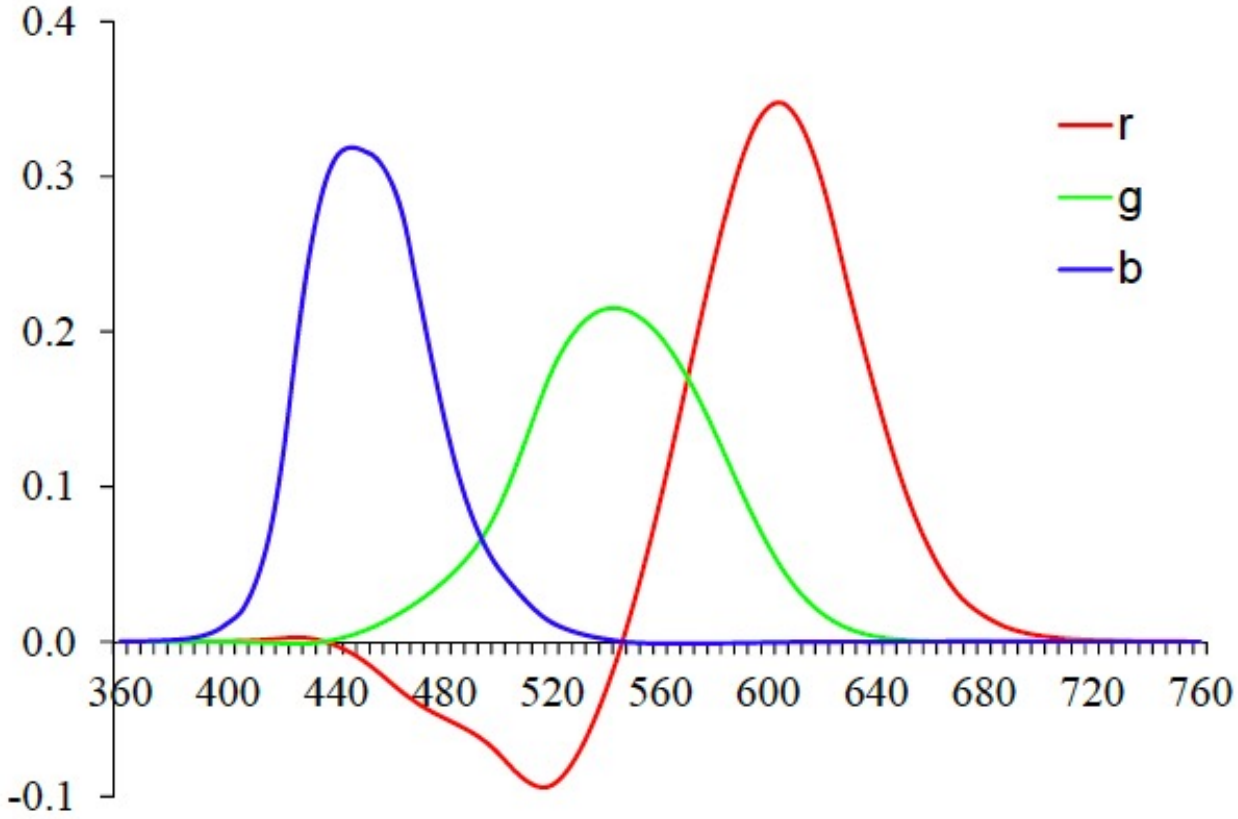


Human perception of color can be studied by asking observers to mix colored lights to match a test light shown in a split field

$$T = w_1 P_1 + w_2 P_2 + w_3 P_3$$

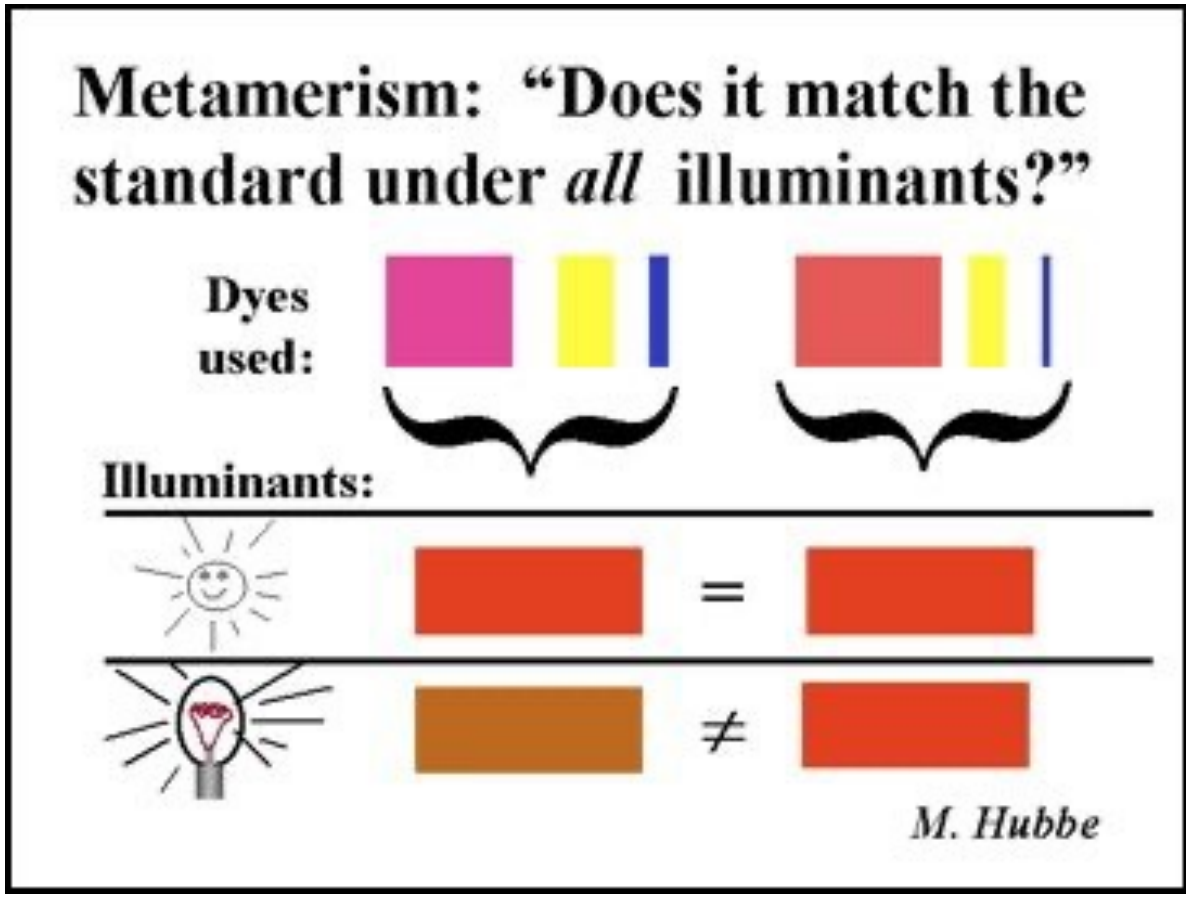
Forsyth & Ponce, Figure 3.2

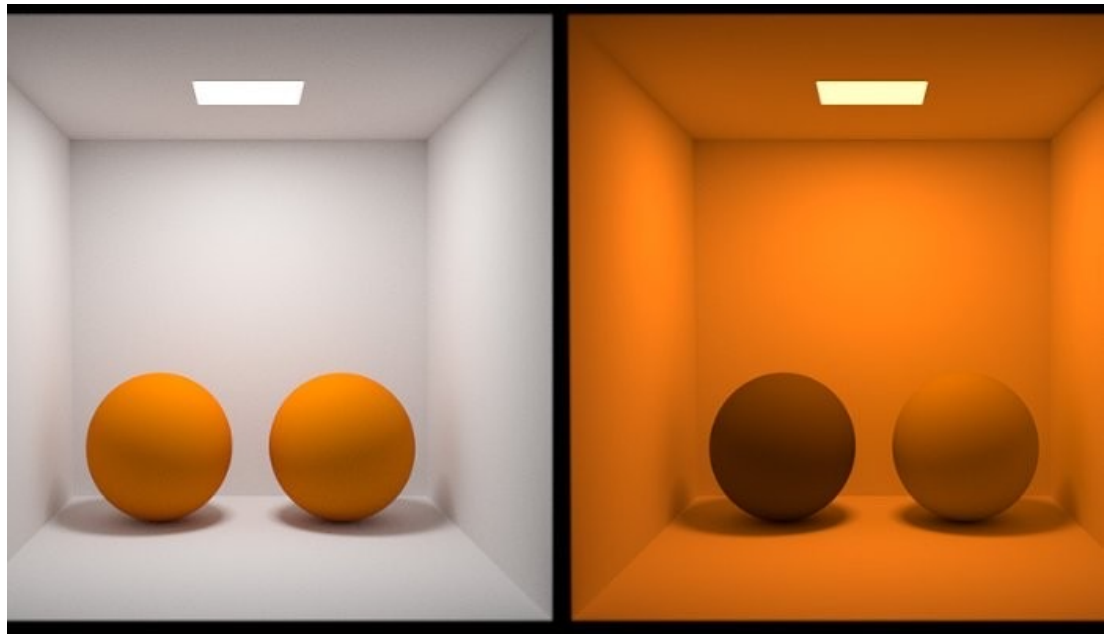
Color Matching by CIE



In the 1930s, the Commission Internationale d'Eclairage (CIE) standardized the RGB representation by performing such color matching experiments using the primary colors of red (700.0nm), green (546.1nm), and blue (435.8nm).

Case Study: Metamer



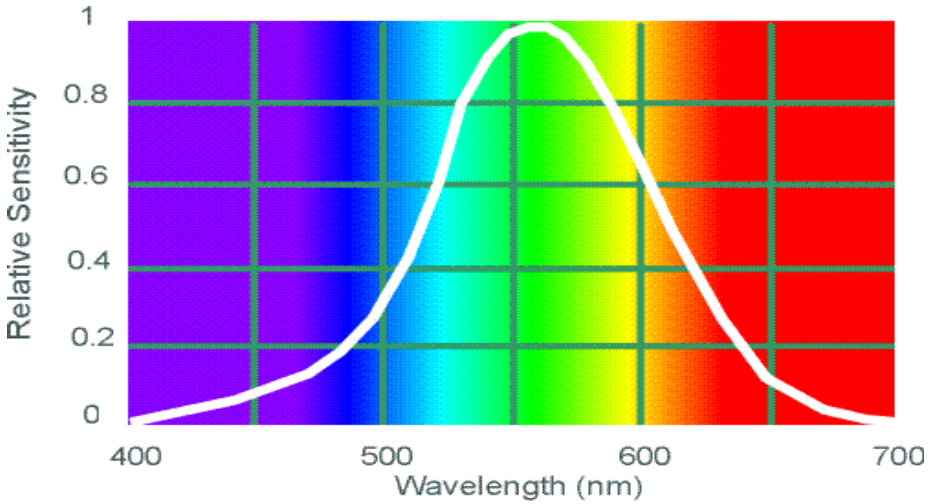


https://www.linkedin.com/pulse/metamerism-how-manage-filippo-busolo

Different spectra, same color response

Case Study: Luminous Efficiency Function





Human Luminance Sensitivity Function

Fluorescence in beer: The one watt laser appears much dimmer than the fluorescence it produces, because the camera, like the human eye, is much more sensitive between 500 and 600 nm than at the laser's 450 nm wavelength.

From wiki

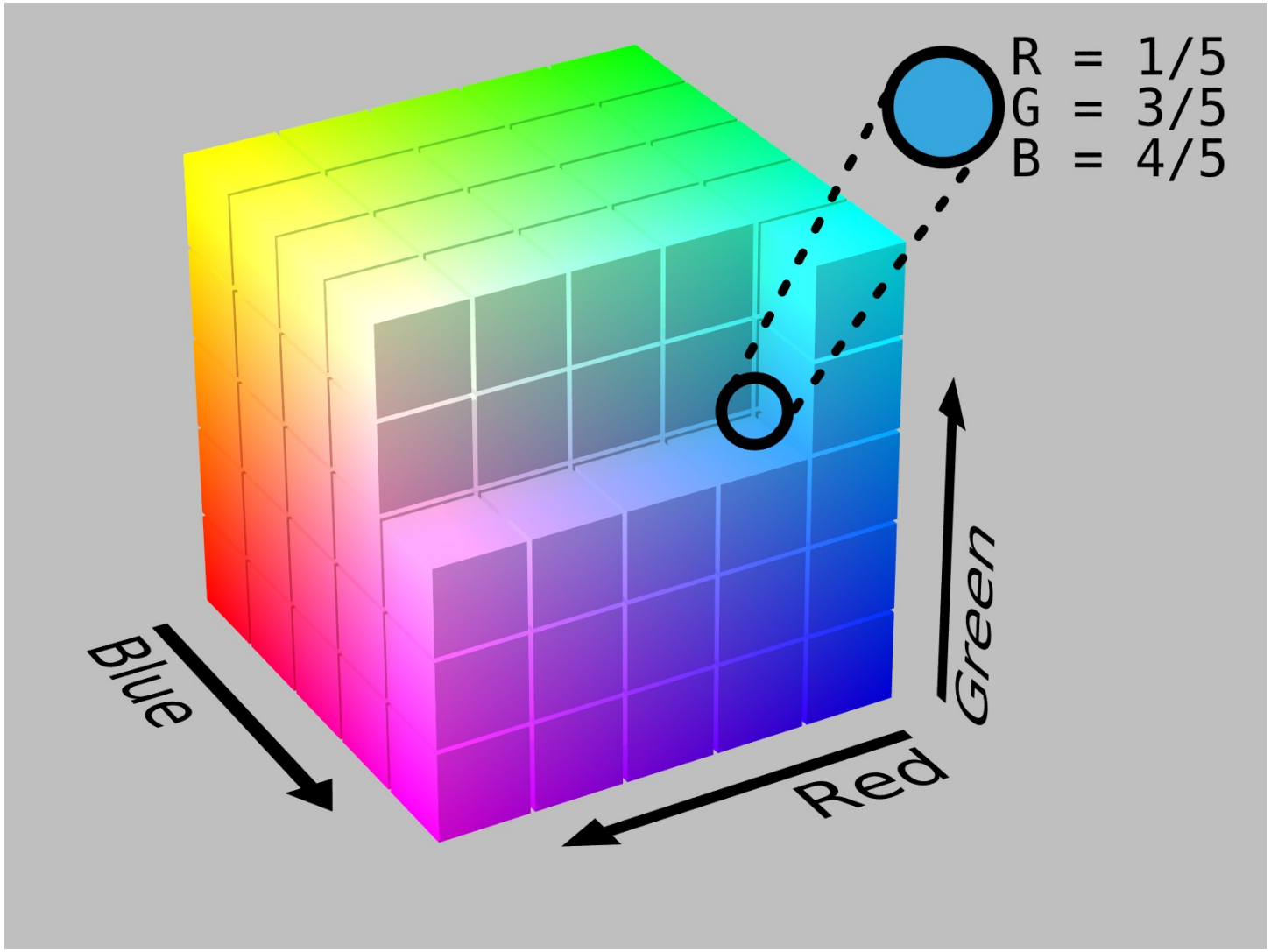
Color Space

Beyond RGB: XYZ and HSV

Standard Color Spaces

- Linear color space
  - Use a set of primaries / color matching functions
  - Example: RGB, XYZ
- Non-linear color space
  - Example: HSV
- Others: CMY, LAB,

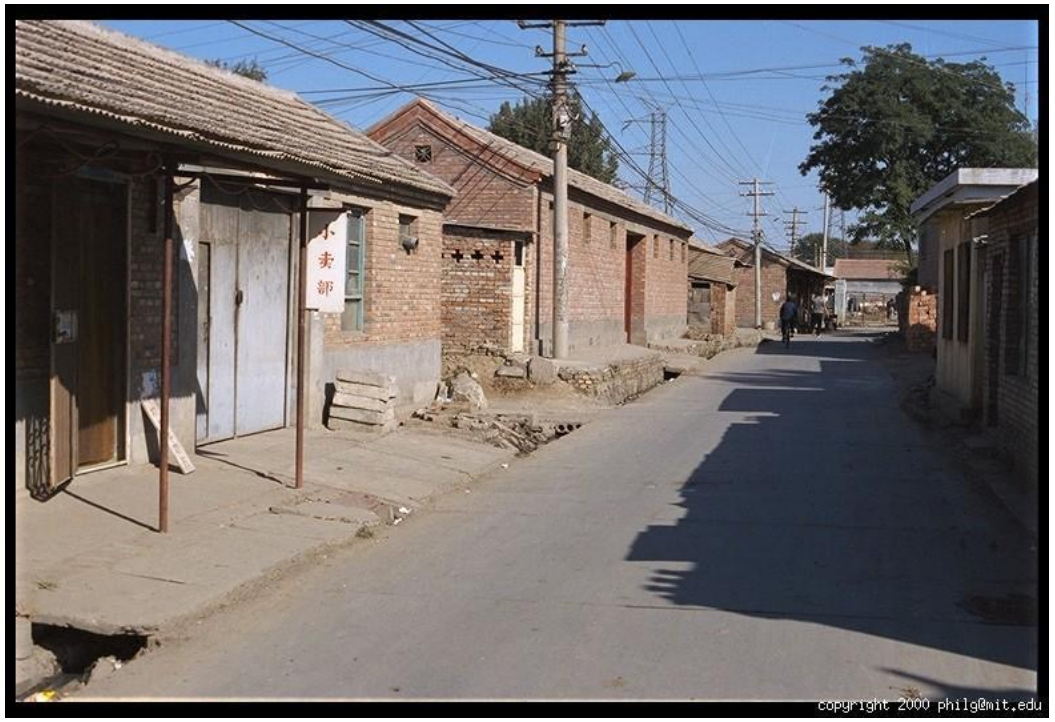
RGB Color Space



Triplet: (R, G, B)

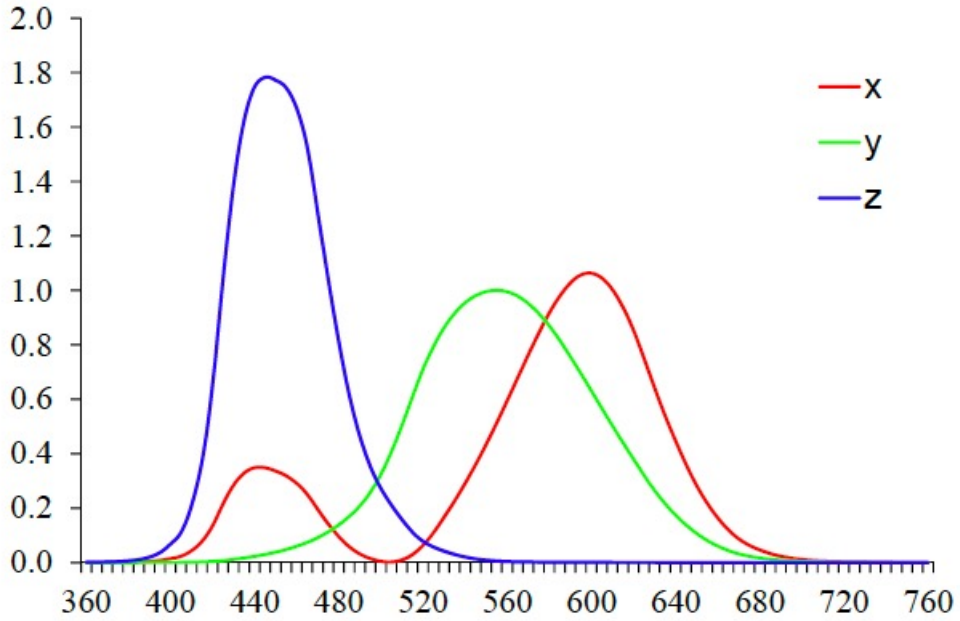
A color image can be a 3D array





HxWx3 array

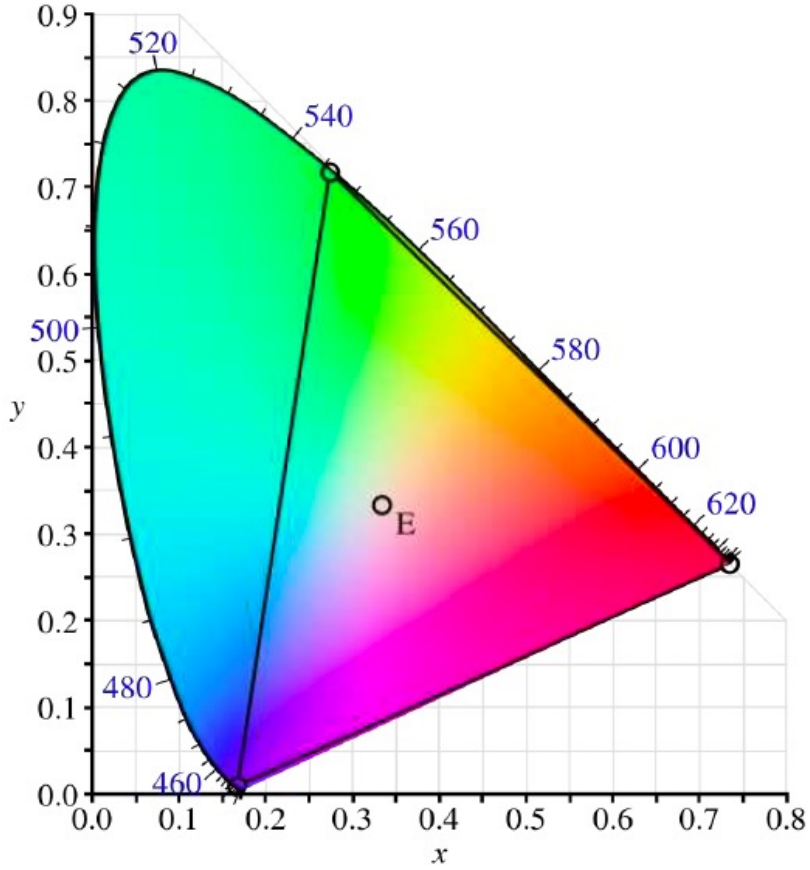
XYZ Color Space (Linear)



Maps the Y axis to the luminance

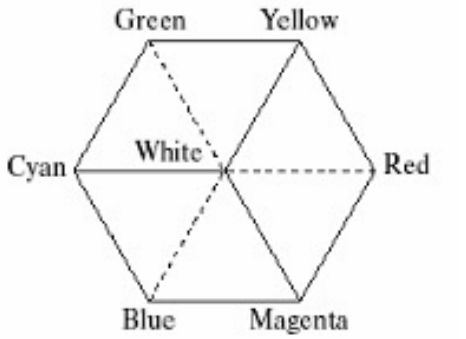
$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \frac{1}{0.17697} \begin{bmatrix} 0.49 & 0.31 & 0.20 \\ 0.17697 & 0.81240 & 0.01063 \\ 0.00 & 0.01 & 0.99 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

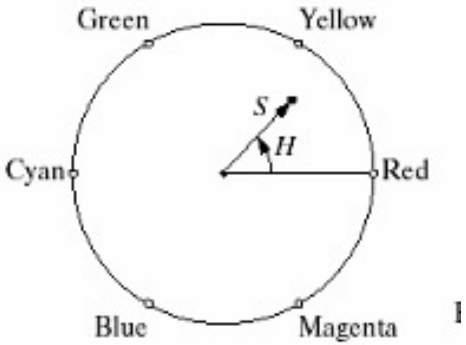
Conversion between RGB and CIE XYZ

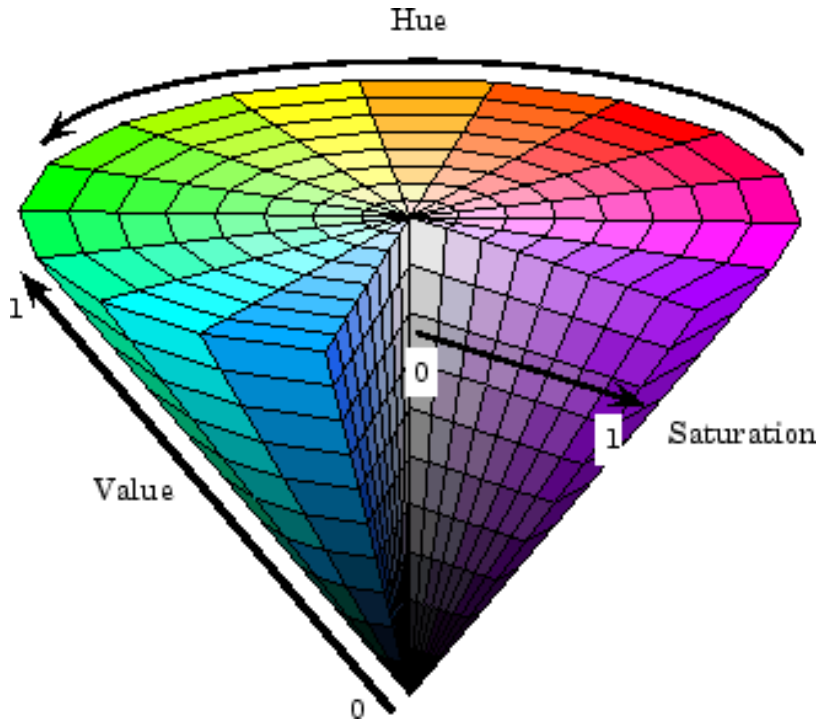


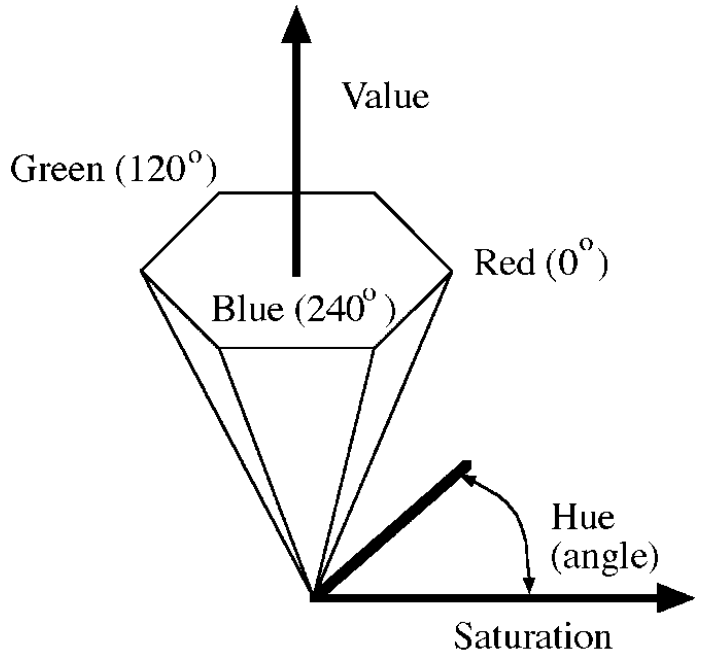
chromaticity diagram of CIE xy color space

HSV Color Space (Nonlinear)



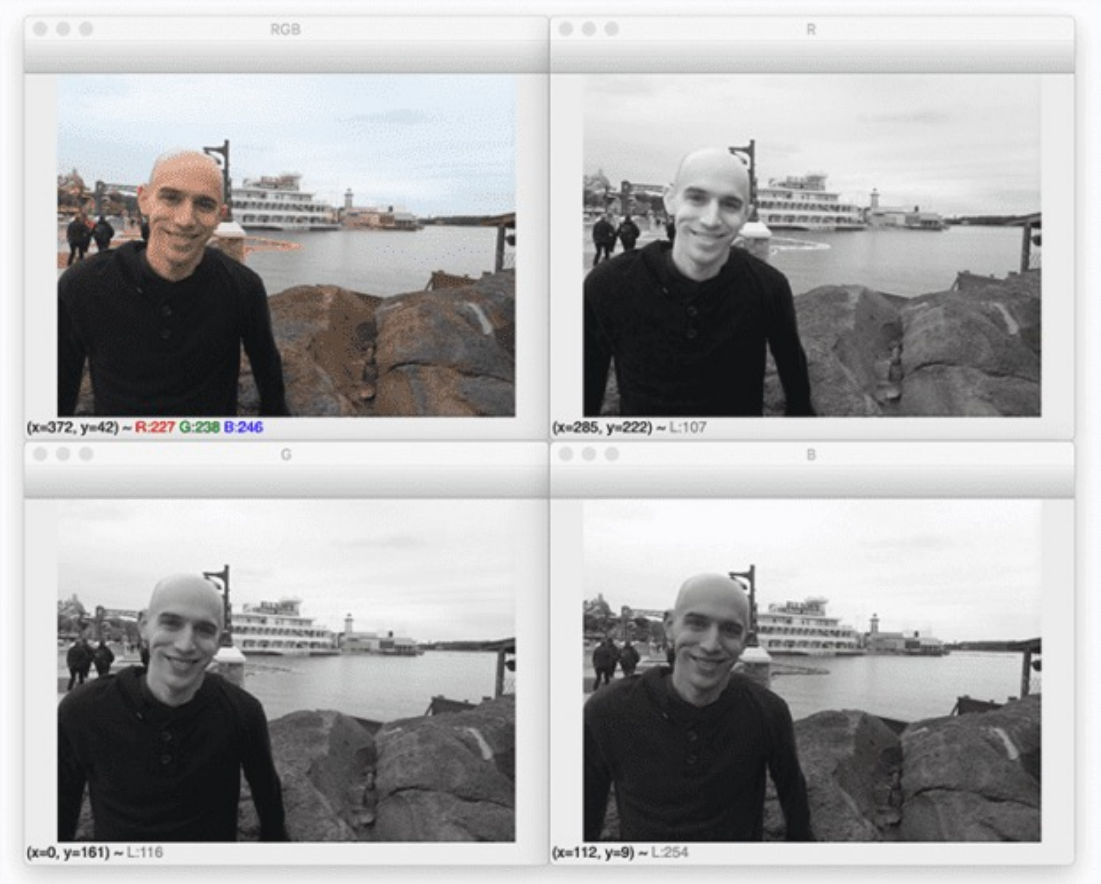


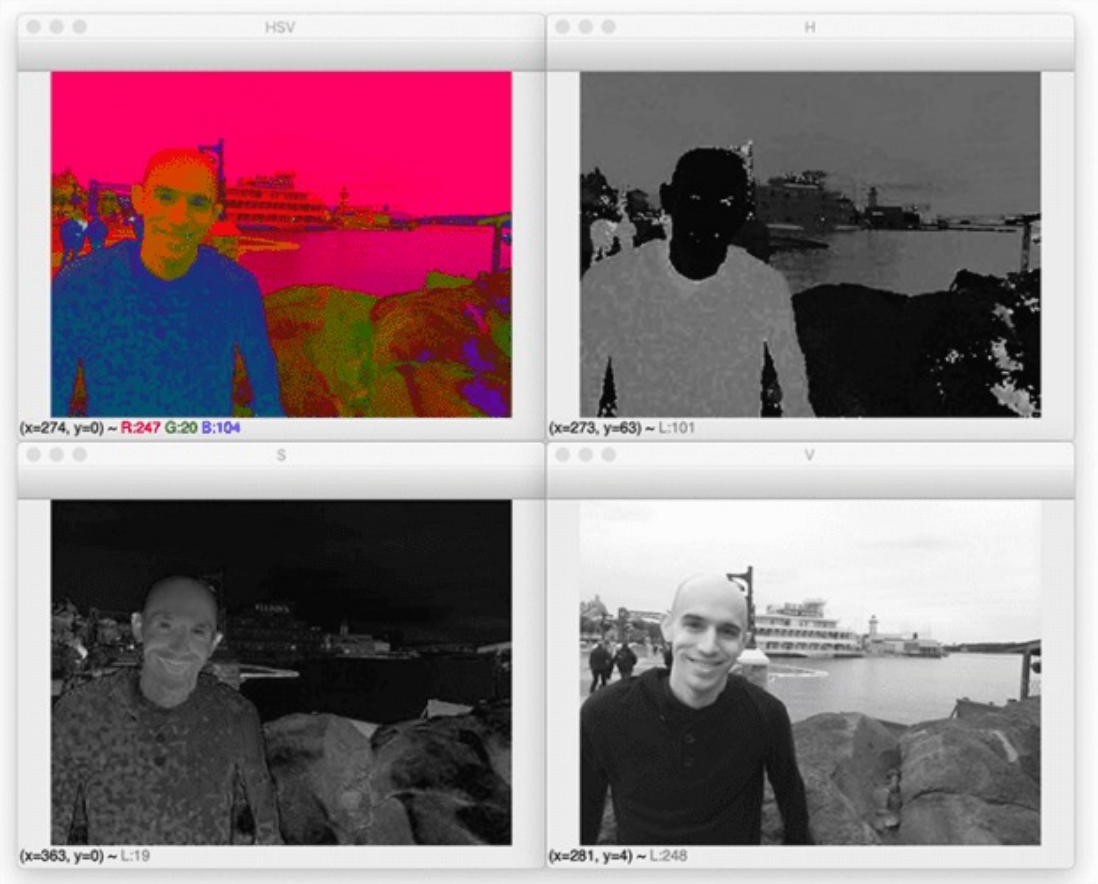




Perceptually meaningful dimensions: Hue (0-360), Saturation (0-1), Value (0-1)

Using OpenCV to Convert Color Spaces





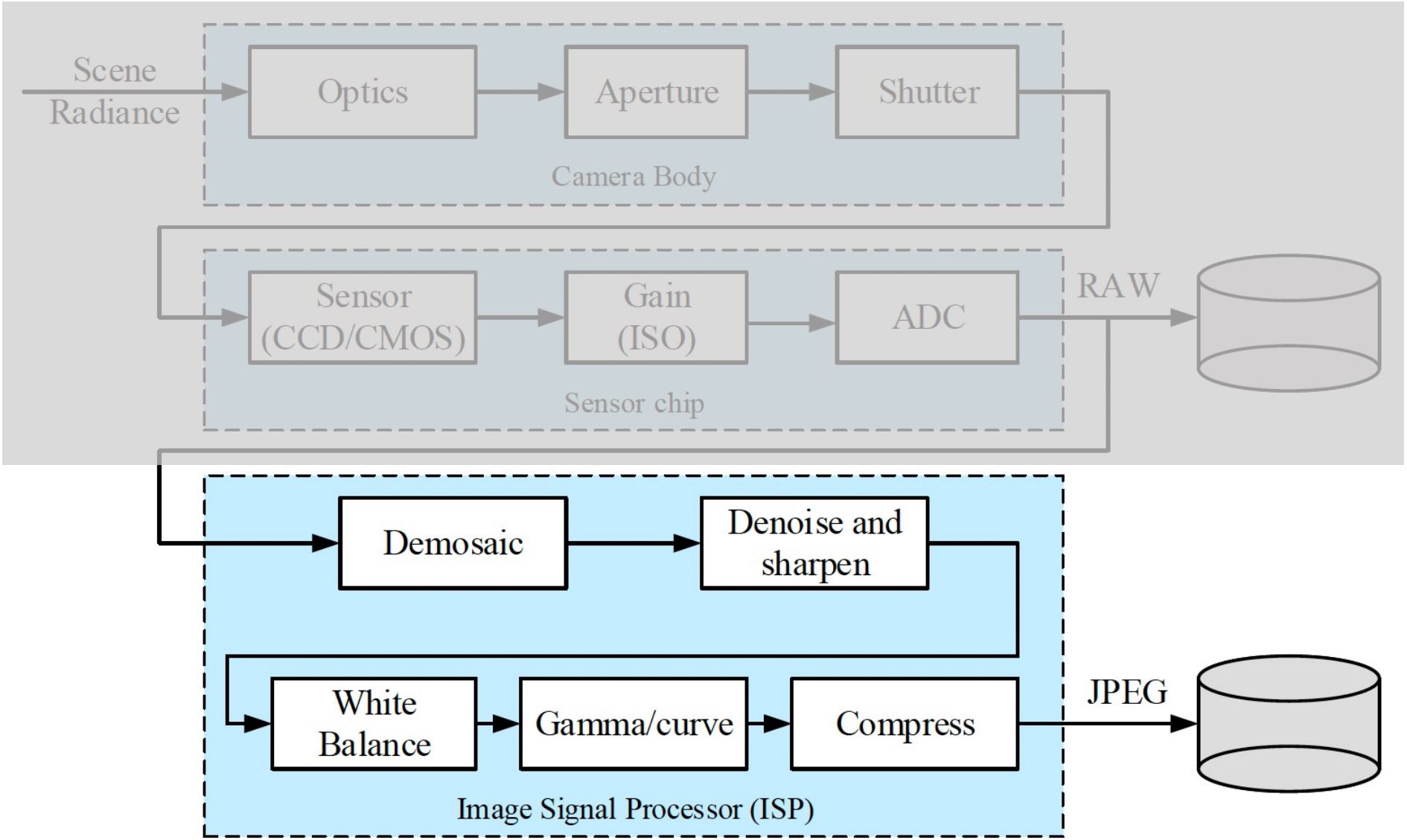

```
cv2.cvtColor(image, cv2.COLOR_BGR2HSV)
```

https://pyimagesearch.com/2021/04/28/opencv-color-spaces-cv2-cvtcolor/

Image Signal Processing

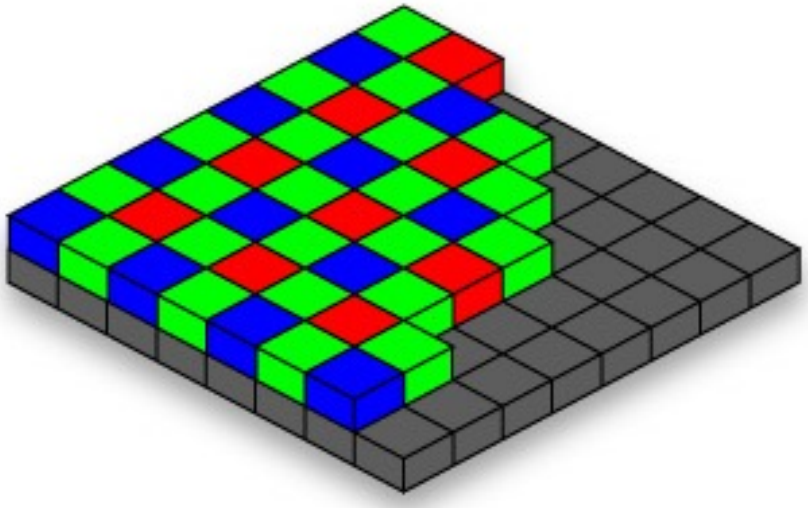
Digital post-processing steps

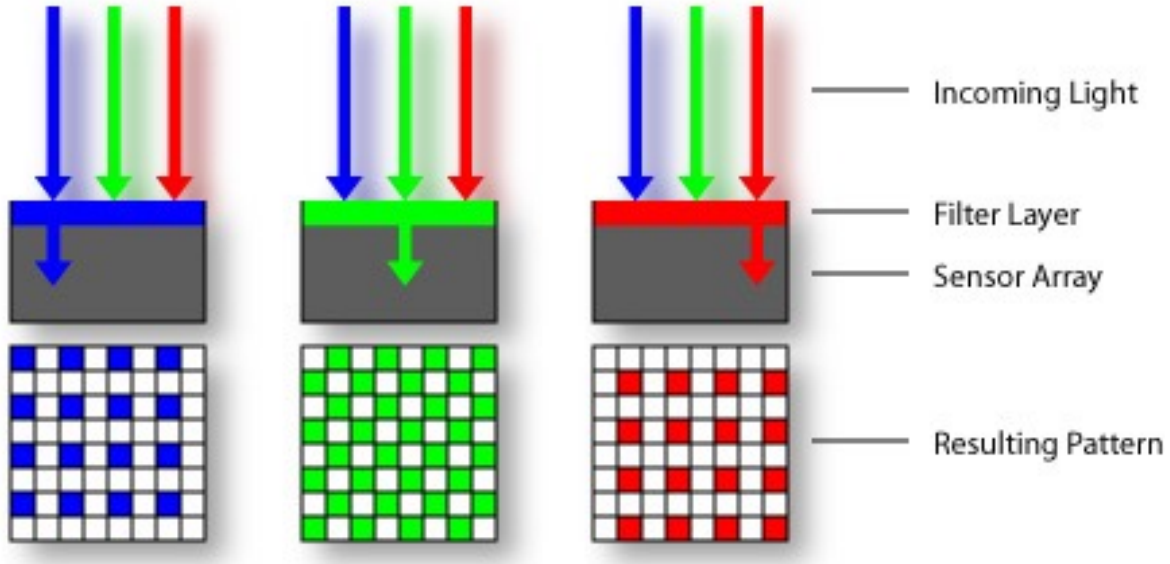
Review: Image Sensing Pipeline (ISP)



Szeliski, Ch 2.3, Figure 2.23

Color Sensing: Bayer Grid





Estimate RGB at each cell from neighboring values

G	R	G	R
B	G	B	G
G	R	G	R
B	G	B	G

Estimated values are lowercase

rGb	Rgb	rGb	Rgb
rgB	rGb	rgB	rGb
rGb	Rgb	rGb	Rgb
rgB	rGb	rgB	rGb

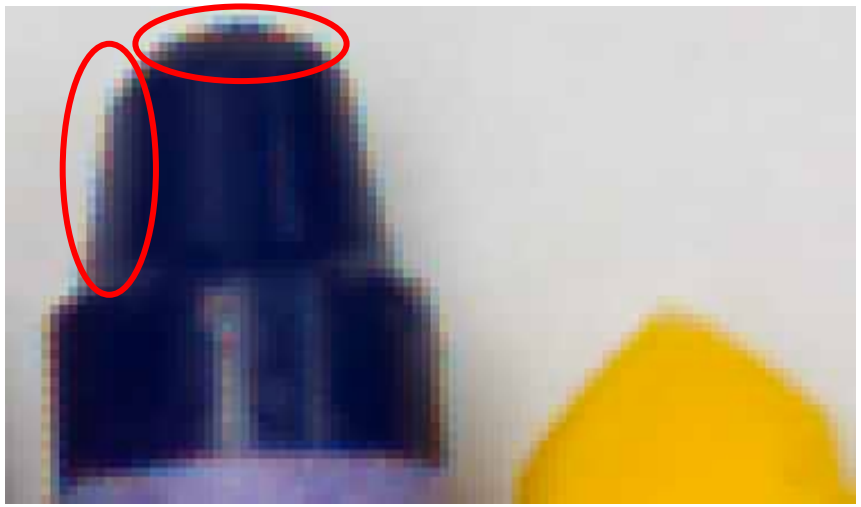
Credit to Steve Seitz

Demosaicing

Demosaicing samples from a color filter array (CFA) into a full-color RGB image by **interpolation**



original full-resolution image

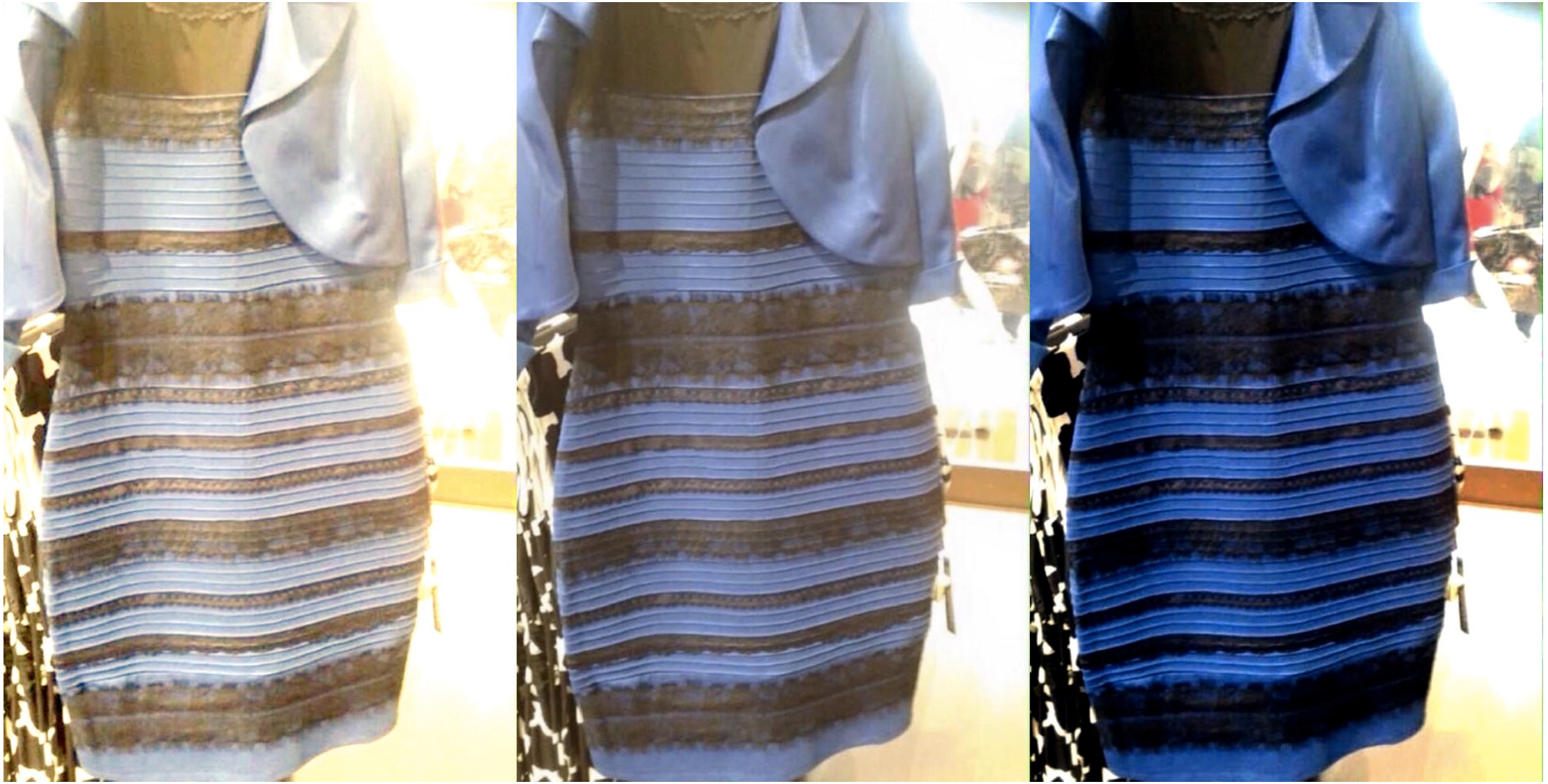


bilinear interpolation results

Artifacts: showing color fringing near the tip of the blue crayon and zippering near its left (vertical) edge;

Szeliski, Figure 10.36

Color Balance

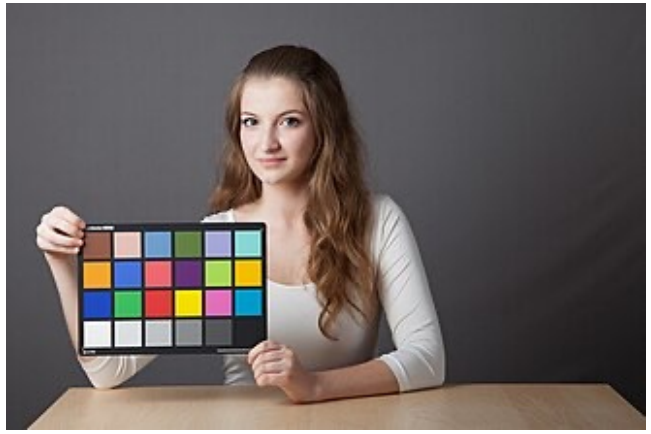


Left: white-balanced as if the dress is white-gold

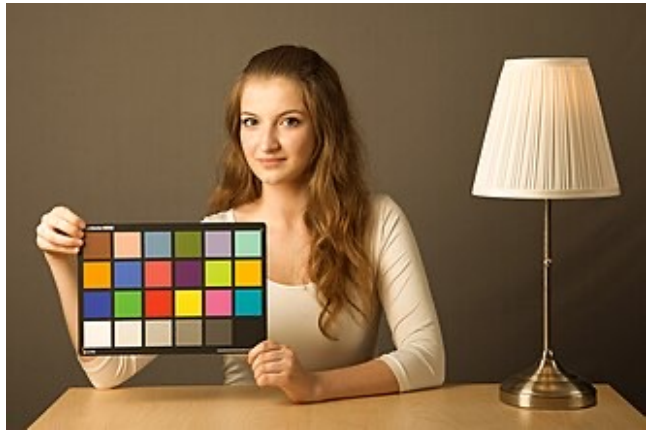
Right: white-balanced as if the dress is blue-black

Color Balance

- **Color balance** is the global adjustment of the intensities of the colors
- An important goal of this adjustment is to render specific colors – particularly neutral colors like white or grey – correctly
- White balance: move the white point of a given image closer to pure white



Neutral light



Warm light

$$\begin{bmatrix} R_{wb} \\ G_{wb} \\ B_{wb} \end{bmatrix} = \begin{bmatrix} g_R & 0 & 0 \\ 0 & g_G & 0 \\ 0 & 0 & g_B \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

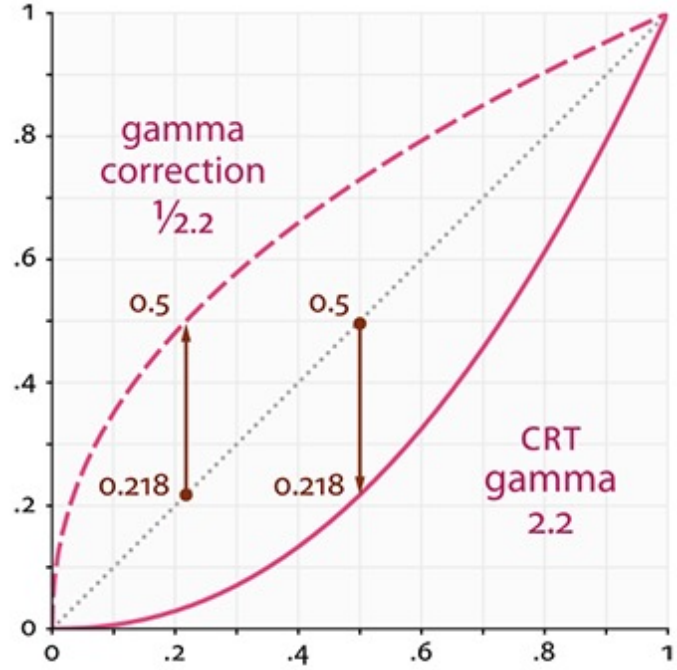
https://en.wikipedia.org/wiki/Color_balance

Gamma Correction



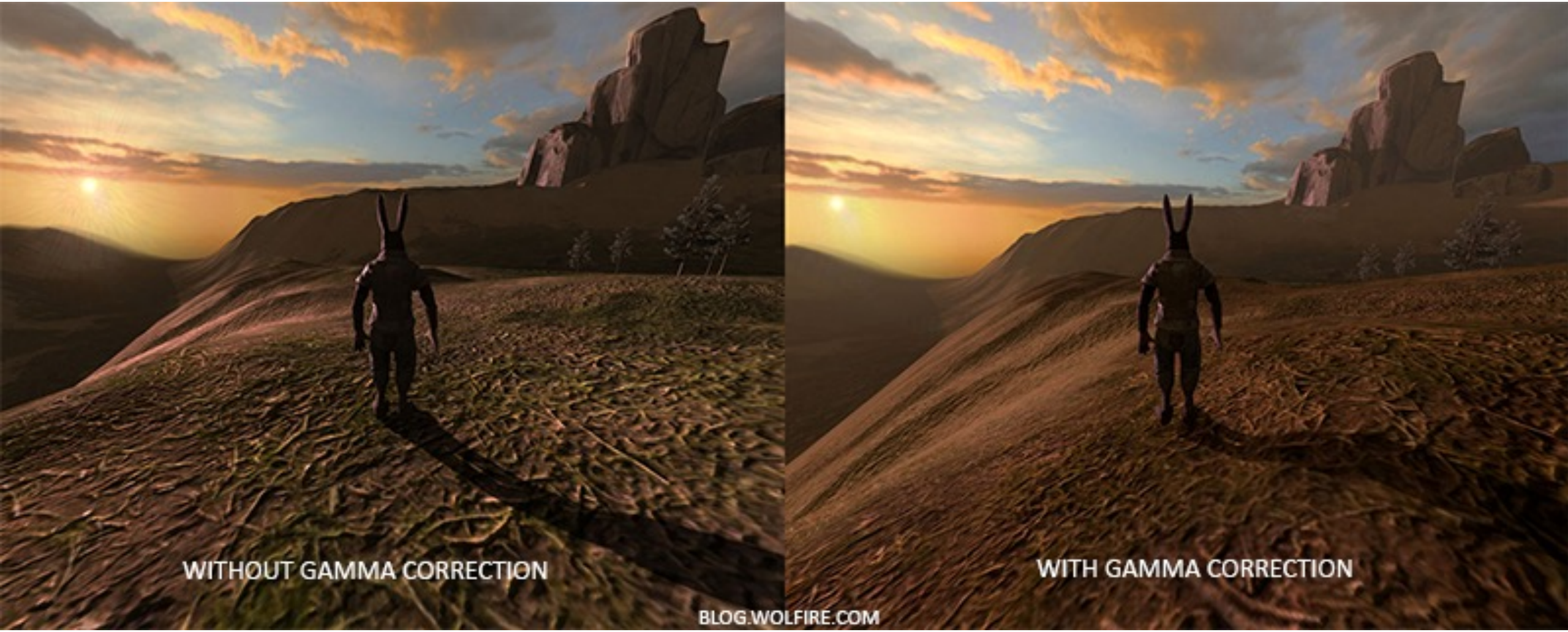
The relationship between the voltage (input to monitor/display) and the resulting brightness was characterized by a number called gamma (about 2.2)

$$B = V^{\gamma}$$



Remember to undo the gamma if your algorithm relies on linear radiance space, e.g., photometric stereo

https://learnopengl.com/Advanced-Lighting/Gamma-Correction


WITHOUT GAMMA CORRECTION
BLOG.WOLFIRE.COM
WITH GAMMA CORRECTION

Applications

Color Perception

- Color/lightness constancy
  - The ability of the human visual system to perceive the intrinsic reflectance properties of the surfaces despite changes in illumination conditions





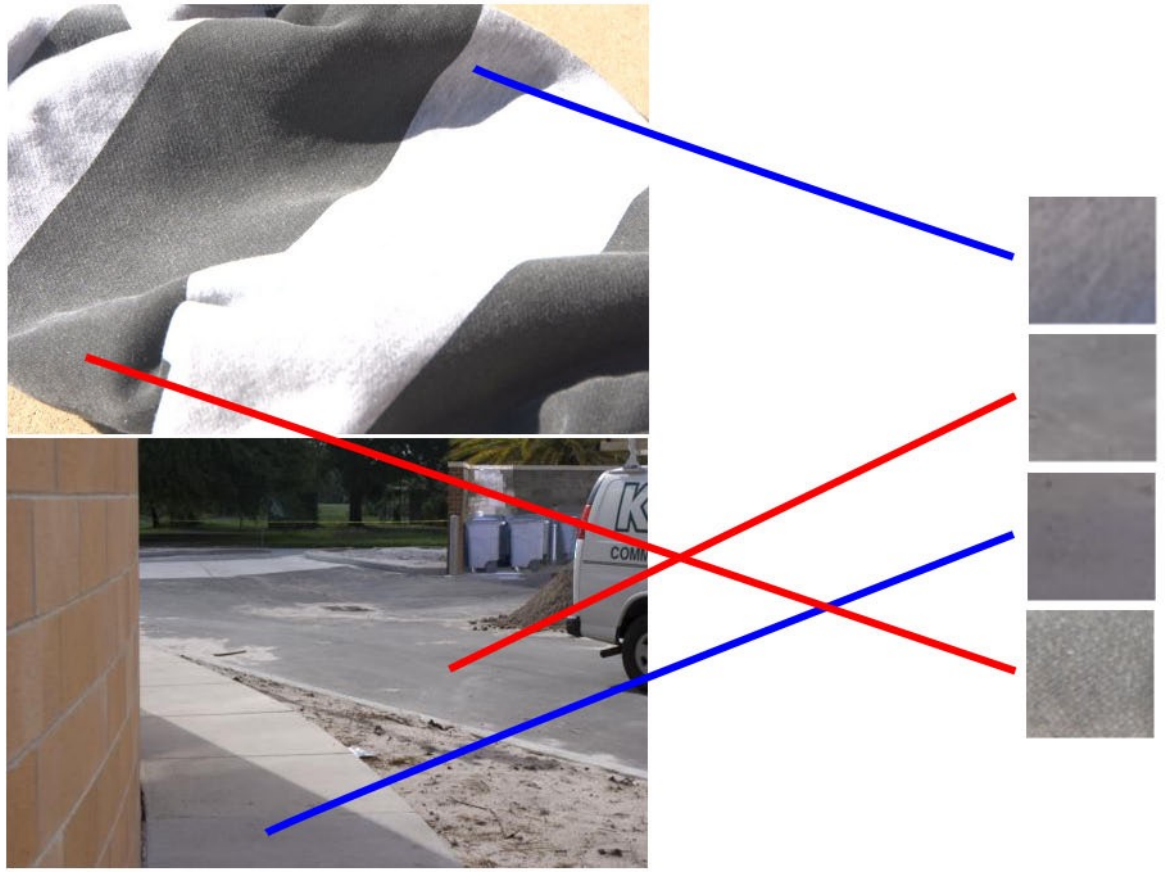




J. S. Sargent, The Daughters of Edward D. Boit, 1882

Color Constancy

- Interpret surface in terms of albedo or “true color”, rather than observed intensity
  - Humans are good at it
  - Computers are not nearly as good



Color Cues in Computer Vision



Image description and matching

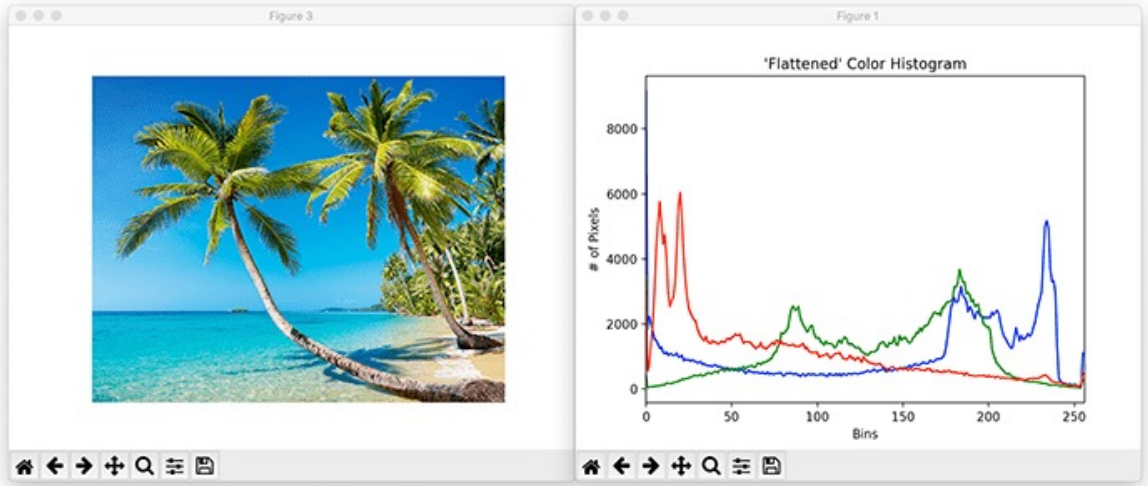
Swain and Ballard, Color Indexing, IJCV 1991



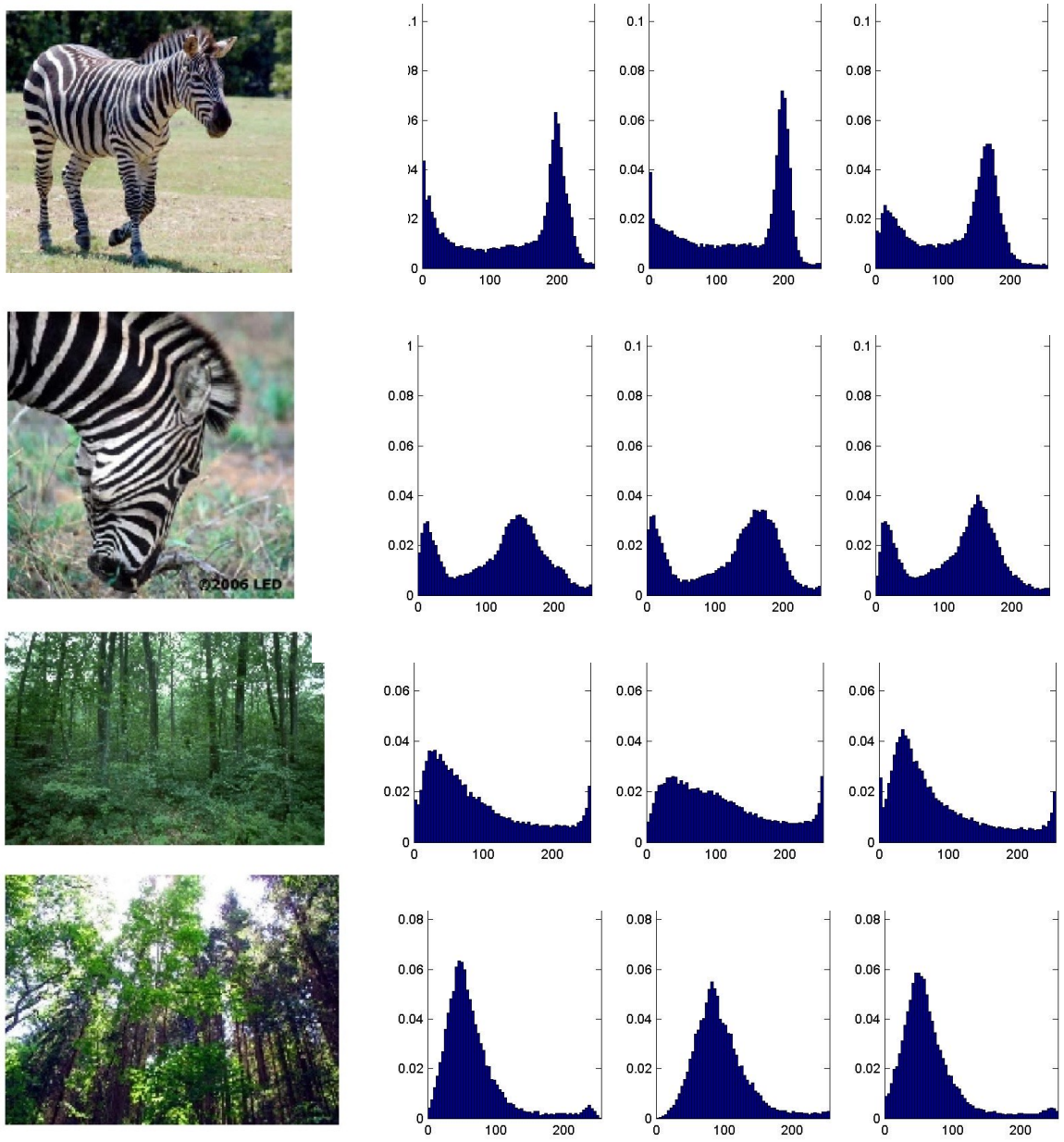
Image segmentation

Carson et al, Blobworld system, 1999

Color Histogram



https://pyimagesearch.com/2021/04/28/opencv-image-histograms-cv2-calchist/



credit to Kristen Grauman

Summary

- Concepts
  - Spectrum
  - Color space: RGB, HSV

- Takeaways
  - Illumination and reflectance determine the spectrum reaching the camera
  - A camera/observer reduces a spectrum to a few weighted measurements
  - Different spectra can produce the same color response
  - Observed color depends on illumination, so color-based CV can be fragile

Reading List

- **Szeliski, *Computer Vision: Algorithms and Applications*, 2e**
  - **2.3.2 Color; 3.1.2 Color Transforms**
  - Optional: 10.3.1 Color Image Demosaicing
- **Forsyth & Ponce, *Computer Vision: A Modern Approach*, 2e**
  - Chapter 3: Color
- **Gonzalez & Woods, *Digital Image Processing*, 4e**
  - Ch. 2: Light & Image Sensing; Ch. 6: Color Image Processing
- M. S. Brown, ICCV 2023 Camera Pipeline Tutorial
  - https://www.eecs.yorku.ca/~mbrown/ICCV2023_Brown.html